

CSE 4621

Machine Learning

Topic: Convolutional Neural Networks

Source: [DeepLearning.AI](https://deeplearning.ai)

Presented by

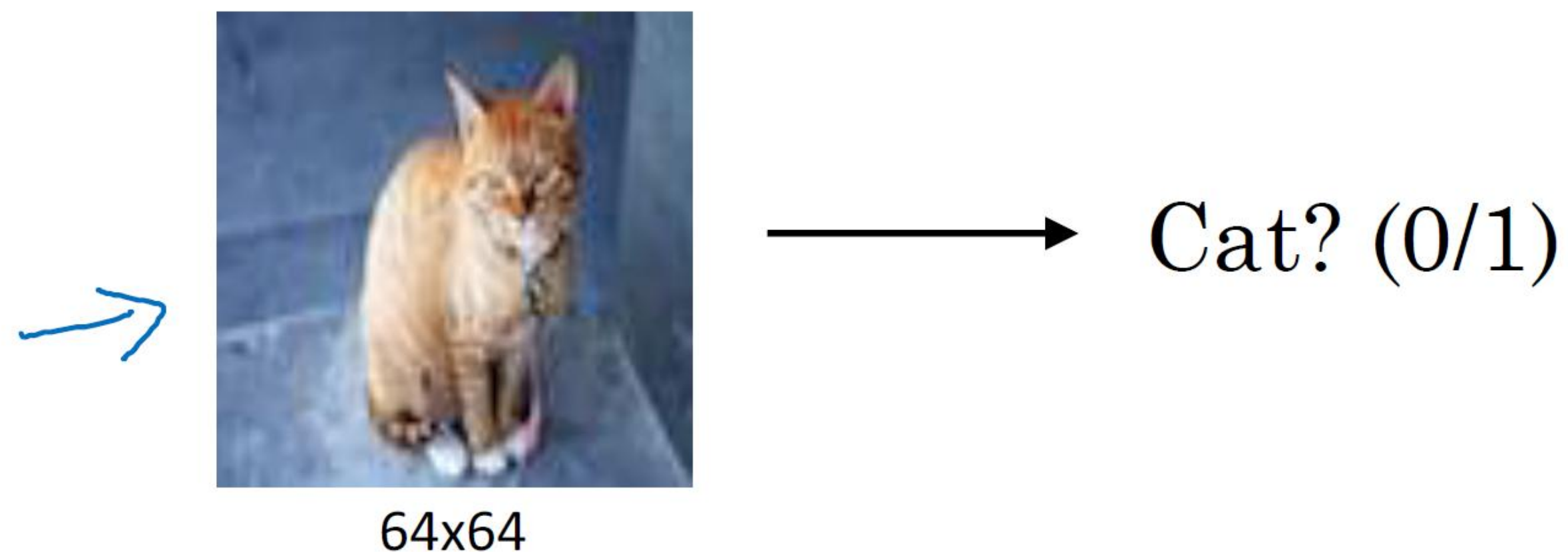
Rafsanjany Kushol

Assistant Professor



Computer Vision Problems

Image Classification



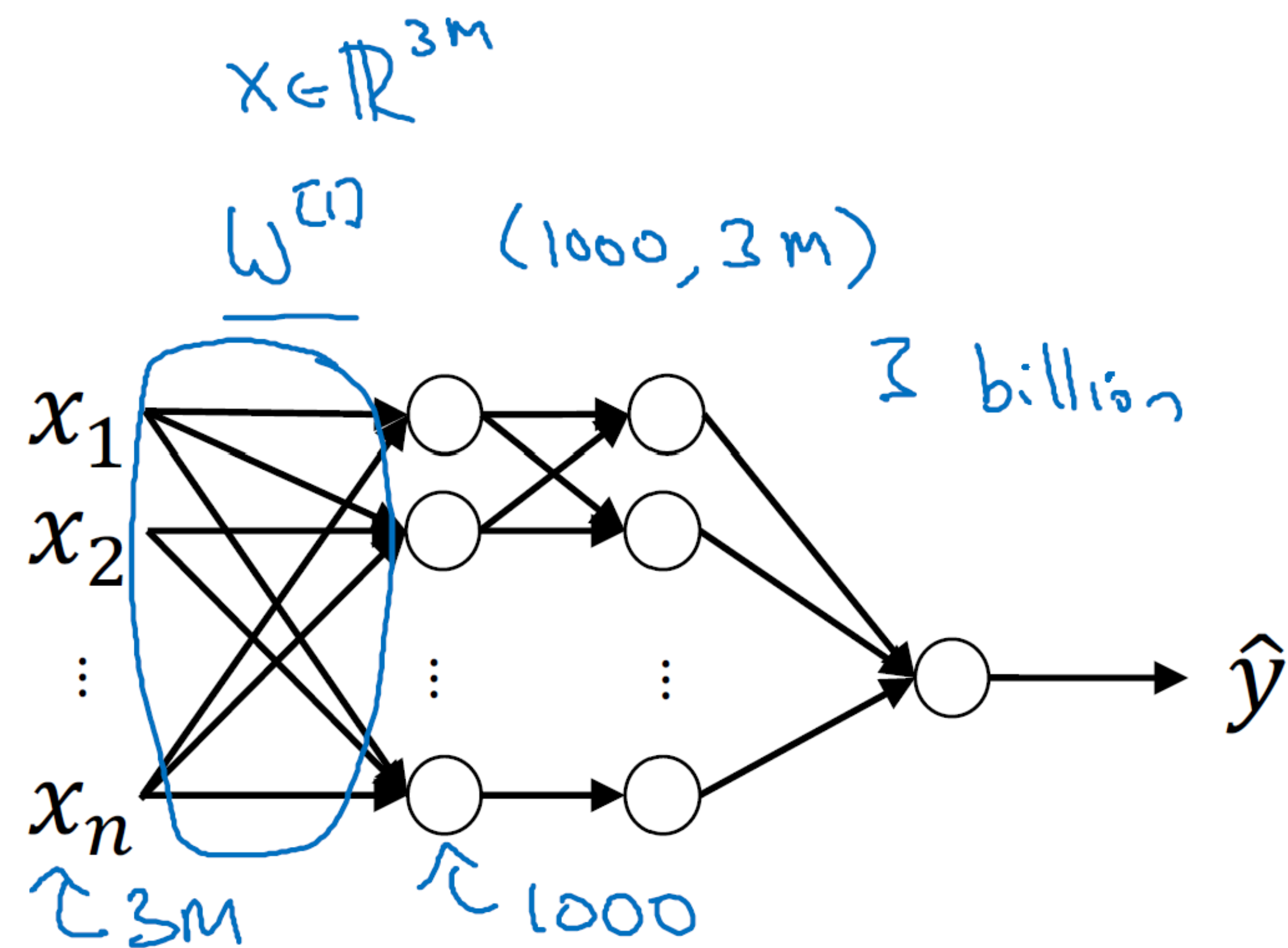
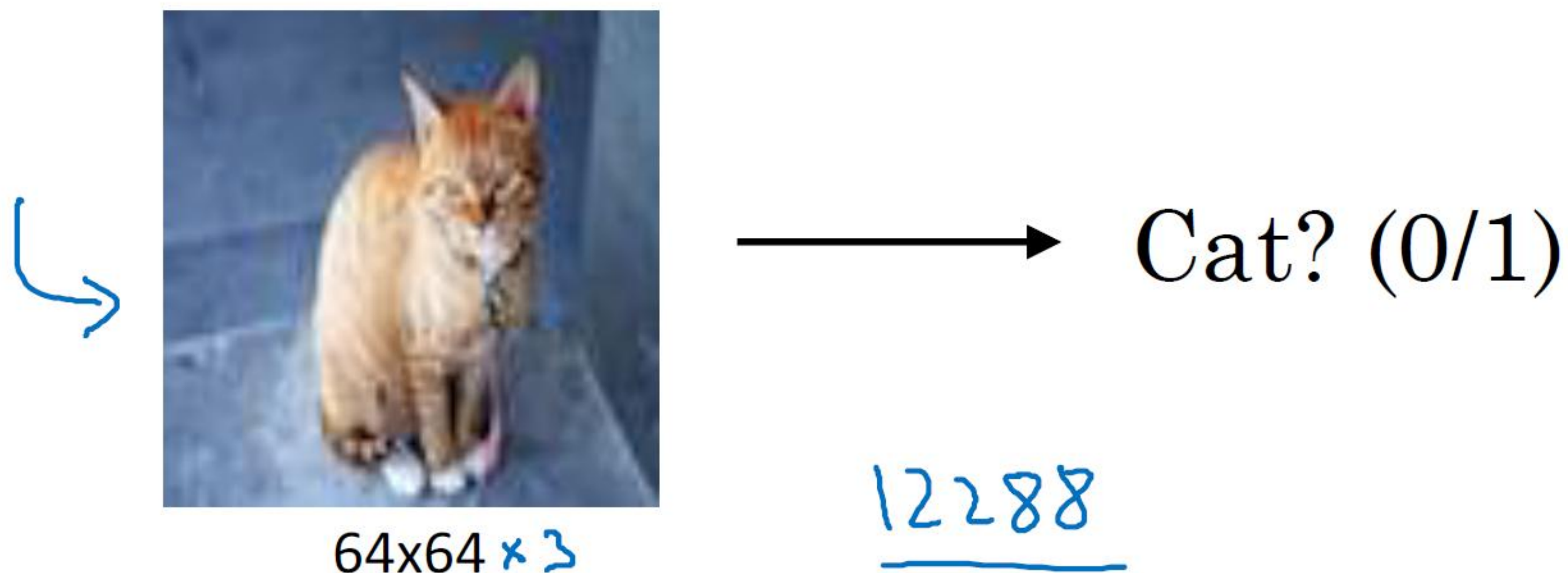
Object detection



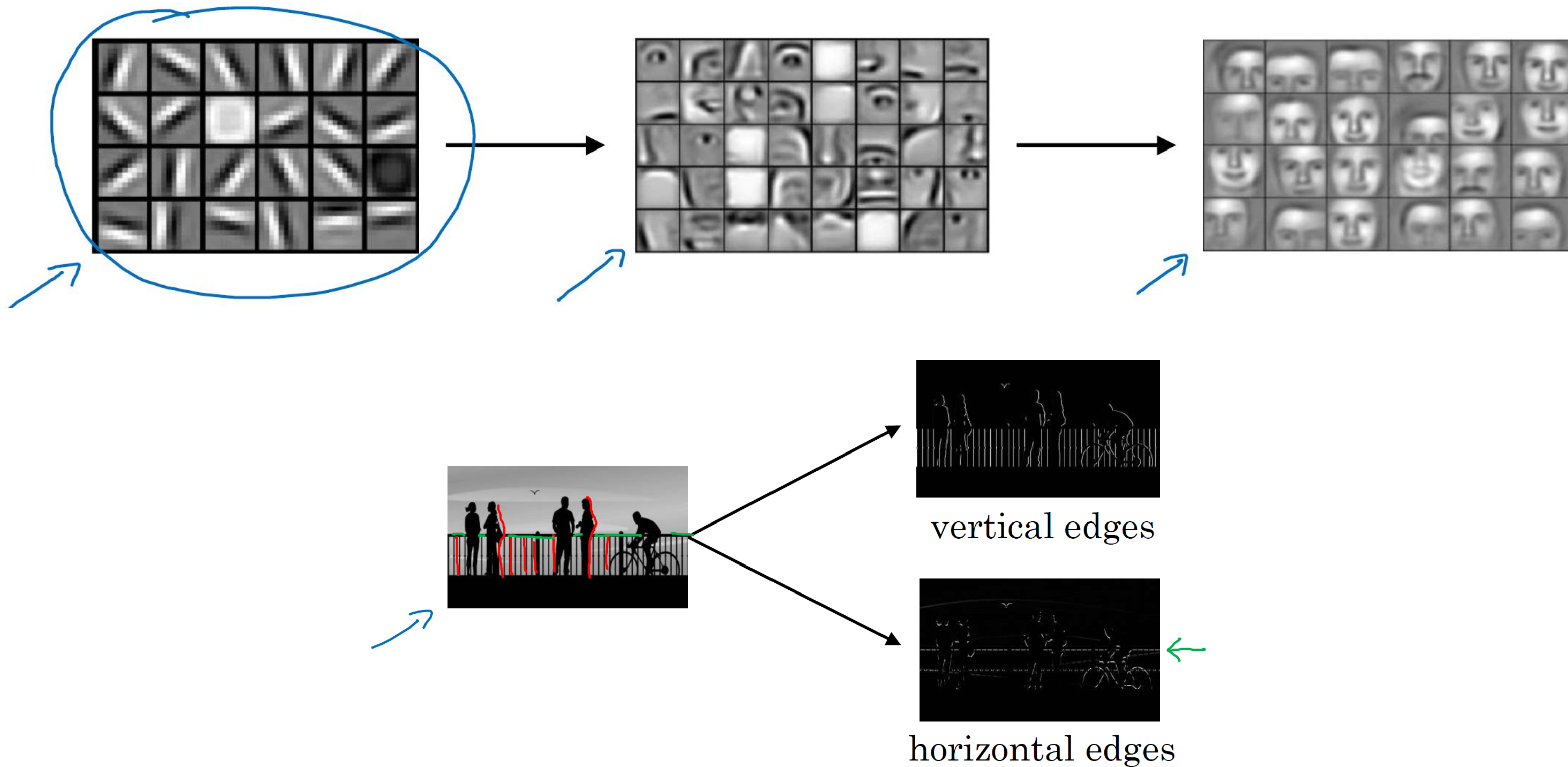
Neural Style Transfer ↙



Deep Learning on large images



Computer Vision Problem



Vertical edge detection

$$3 \times 1 + 1 \times 1 + 2 \times 1 + 0 \times 0 + 5 \times 0 + 7 \times 0 + 1 \times -1 + 8 \times -1 + 2 \times -1 = -5$$

3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6

"convolution"

*

1	0	-1
1	0	-1
1	0	-1

3x3
filter

=

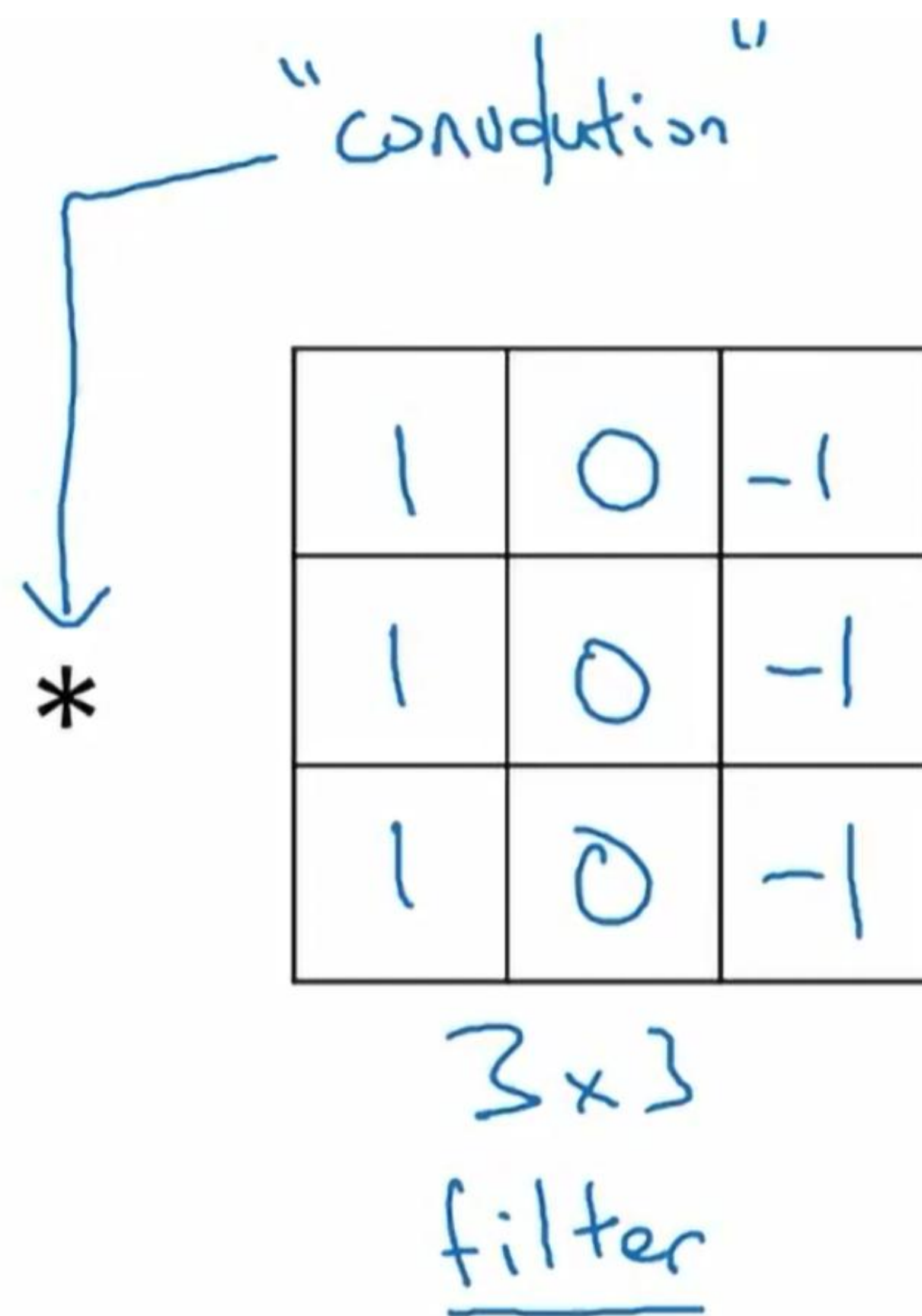
-5			

4x4

Vertical edge detection

3	0 ¹	1 ⁰	2 ⁻¹	7	4
1	5 ¹	8 ⁰	9 ⁻¹	3	1
2	7 ¹	2 ⁰	5 ⁻¹	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6



=

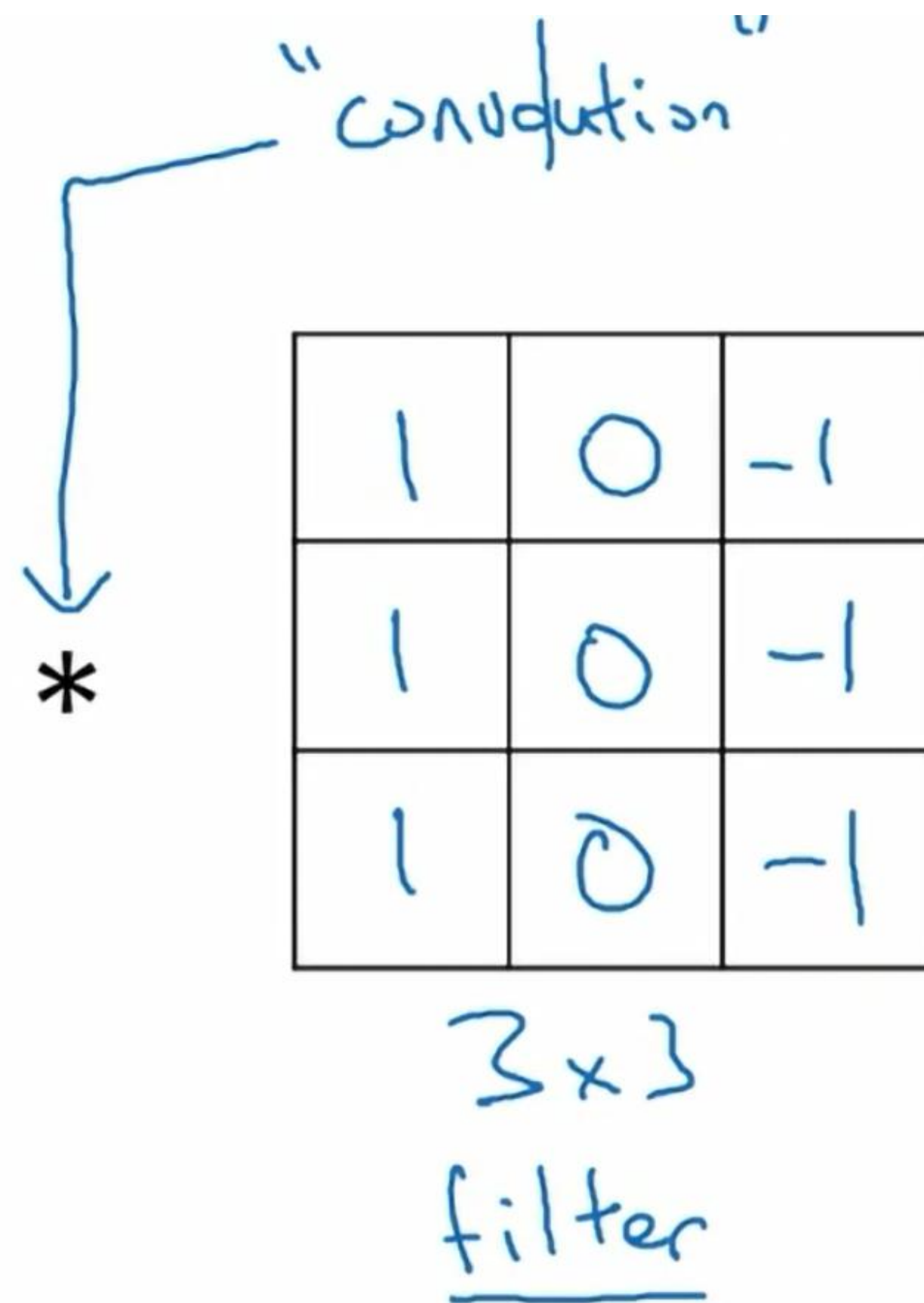
-5			

4x4

Vertical edge detection

3	<u>0</u>	<u>1</u>	<u>2</u>	<u>7</u>	<u>4</u>
¹ <u>1</u>	⁰ <u>5</u>	⁻¹ <u>8</u>	<u>9</u>	<u>3</u>	<u>1</u>
¹ <u>2</u>	⁰ <u>7</u>	⁻¹ <u>2</u>	<u>5</u>	<u>1</u>	<u>3</u>
¹ <u>0</u>	⁰ <u>1</u>	⁻¹ <u>3</u>	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6



=

<u>-5</u>	<u>-4</u>	0	<u>8</u>

4x4

Vertical edge detection

3	<u>0</u>	<u>1</u>	<u>2</u>	<u>7</u>	<u>4</u>
1	<u>5</u>	<u>8</u>	<u>9</u>	<u>3</u>	<u>1</u>
2	<u>7</u>	<u>2</u>	<u>5</u>	<u>1</u>	<u>3</u>
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

6x6

"convolution"

*

1	0	-1
1	0	-1
1	0	-1

3x3
filter

=

<u>-5</u>	<u>-4</u>	0	<u>8</u>
<u>-10</u>	-2	2	3
0	-2	-4	-7
-3	-2	-3	-16

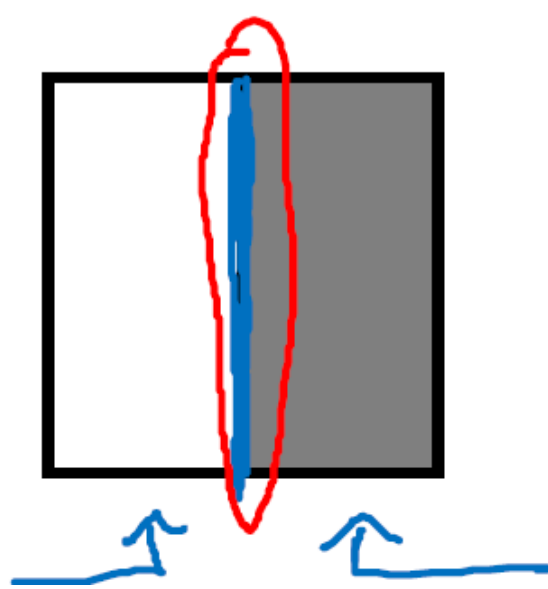
4x4

Vertical edge detection

↓

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0
10	<u>10</u>	<u>10</u>	<u>0</u>	0	0

6x6



*

↓

1	0	-1
1	0	-1
1	0	-1

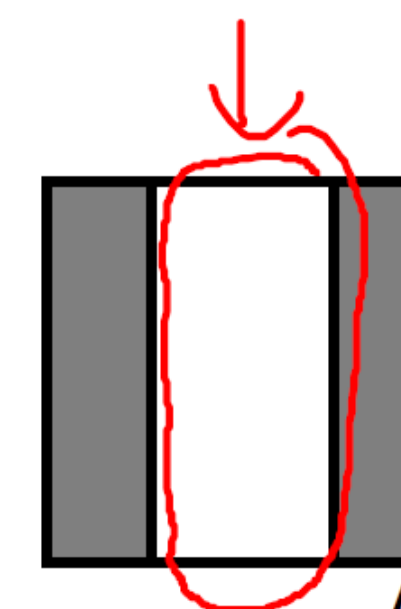
3x3

=

↓

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0

4x4



Andrew Ng

Vertical edge detection examples

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0



*

1	0	-1
1	0	-1
1	0	-1



=

0	30	30	0
0	30	30	0
0	30	30	0
0	30	30	0



0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10



*

1	0	-1
1	0	-1
1	0	-1



=

0	-30	-30	0
0	-30	-30	0
0	-30	-30	0
0	-30	-30	0



Vertical and Horizontal Edge Detection

→

1	0	-1
1	0	-1
1	0	-1

Vertical

→

1	1	1
0	0	0
-1	-1	-1

Horizontal

10	10	10	0	0	0
10	10	10	0	0	0
10	10	10	0	0	0
0	0	0	10	10	10
0	0	0	10	10	10
0	0	0	10	10	10

6x6

*

1	1	1
0	0	0
-1	-1	-1

=

0	0	0	0
30	10	-10	-30
30	10	-10	-30
0	0	0	0

Learning to detect edges

1	0	-1
1	0	-1
1	0	-1



1	0	-1
2	0	-2
1	0	-1

Sobel filter



3	0	-3
10	0	-10
3	0	-3

Scharr filter



3	0	1	2	7	4
1	5	8	9	3	1
2	7	2	5	1	3
0	1	3	1	7	8
4	2	1	6	2	8
2	4	5	2	3	9

convolution
*

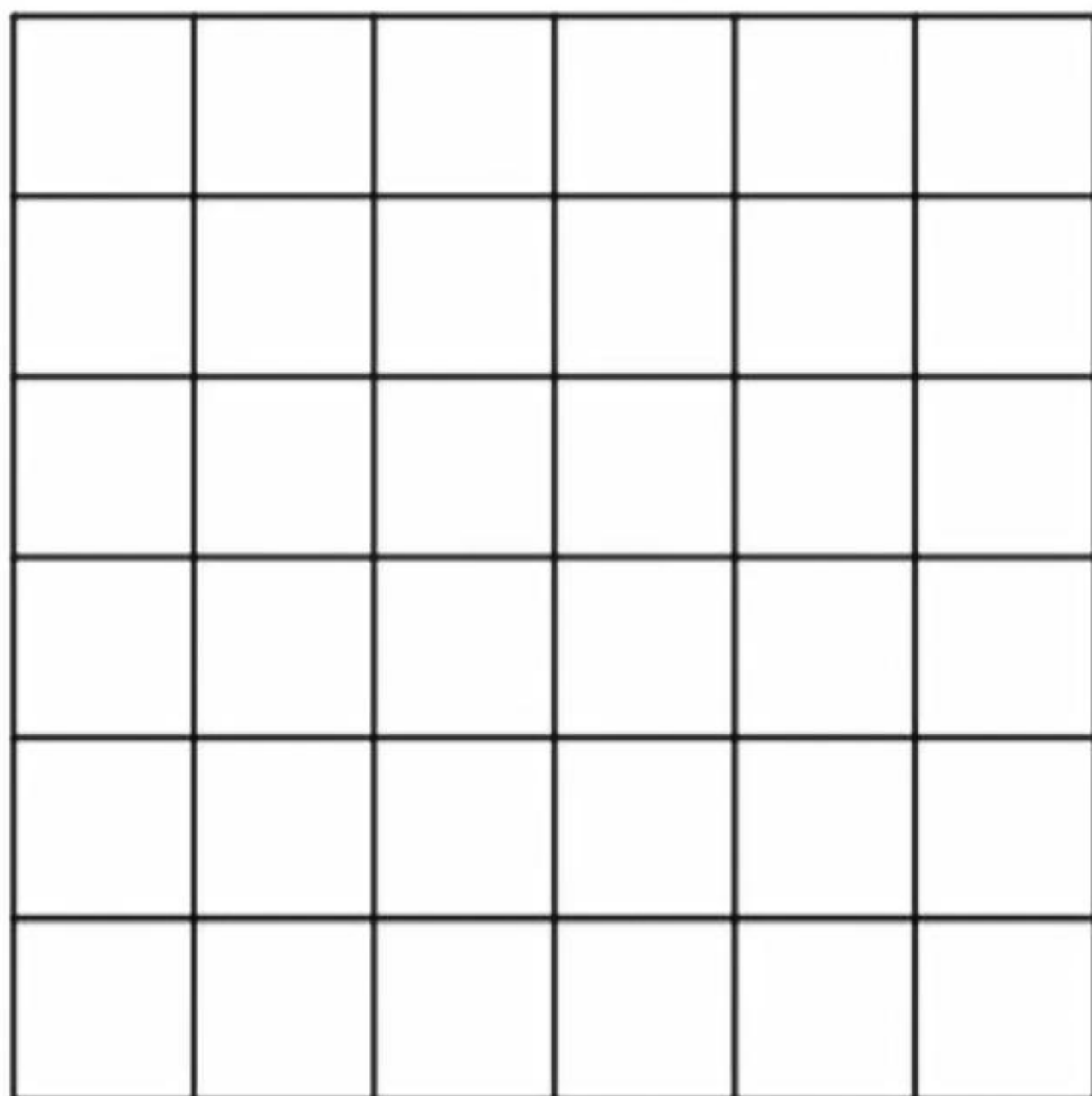
W_1	W_2	W_3
W_4	W_5	W_6
W_7	W_8	W_9

3x3

=

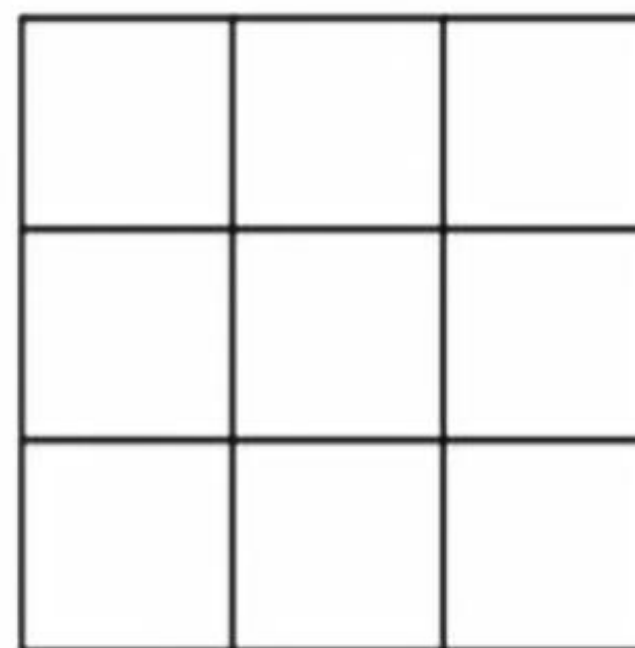
45°
70°
73°

Padding



6×6
 $n \times n$

*



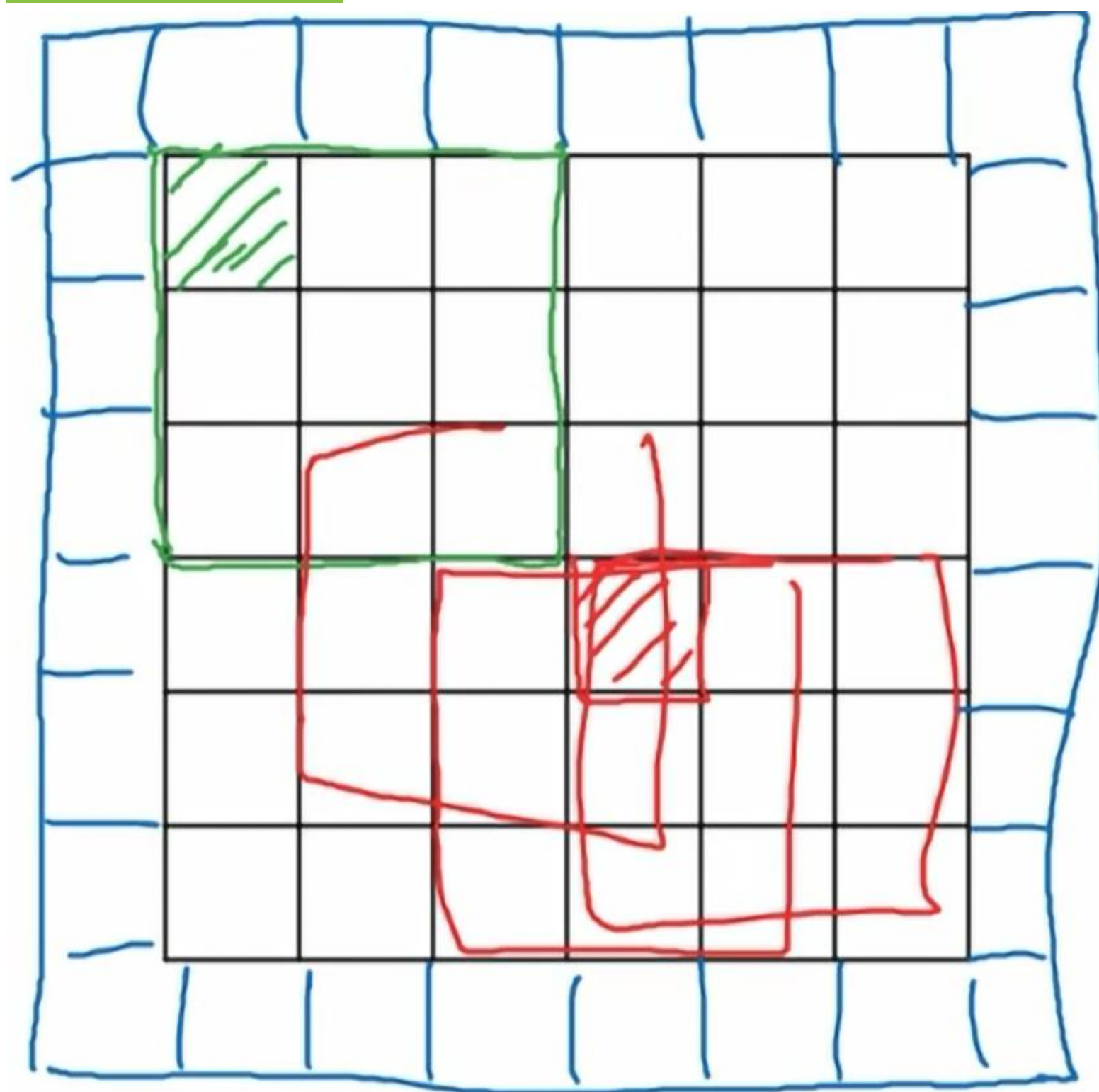
3×3
 $f \times f$

=

$n - f + 1 \times n - f + 1$
 $6 - 3 + 1 = 4$

→ 4×4

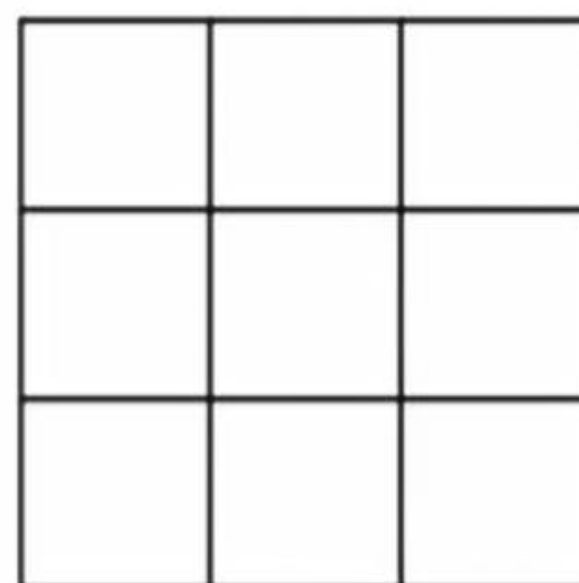
Padding



$$\begin{array}{c} 6 \times 6 \\ \hline n \times n \end{array} \rightarrow 8 \times 8$$

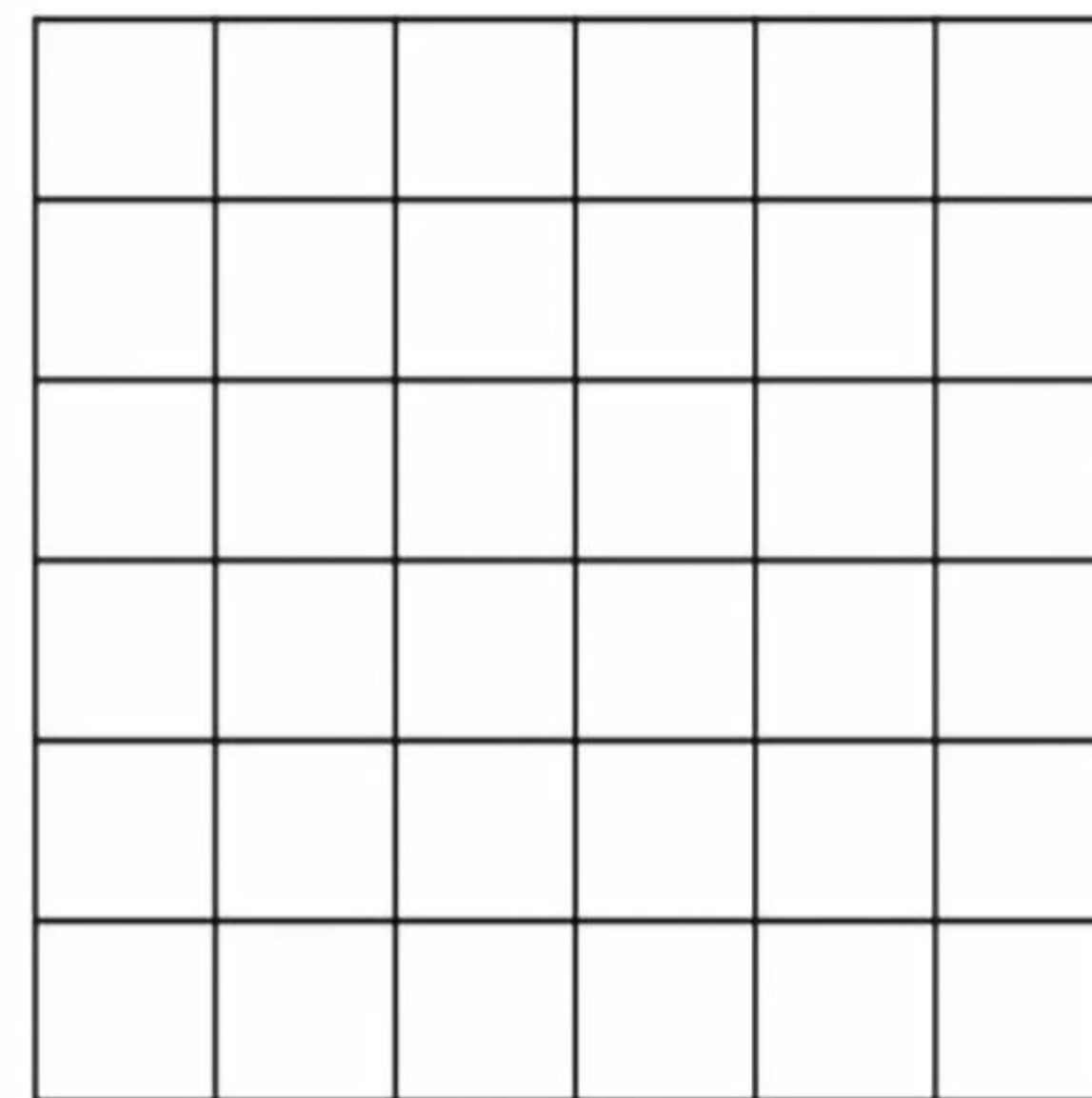
$$p = \text{padding} = 1$$

*



$$\begin{array}{c} 3 \times 3 \\ f \times f \end{array}$$

=



$$6 \times 6$$

$$\begin{array}{c} n - f + 1 \times n - f + 1 \\ 6 - 3 + 1 = 4 \end{array}$$

$$\rightarrow \underline{\underline{4 \times 4}}$$

$$\begin{array}{c} n + 2p - f + 1 \times n + 2p - f + 1 \\ 6 + 2 - 3 + 1 \times \underline{\quad} = 6 \times 6 \end{array}$$

Valid and Same convolutions

→ no padding

“Valid”: $n \times n$ $\ast$ $f \times f$ $\rightarrow \frac{n-f+1}{1} \times n-f+1$
 6×6 $\ast$ 3×3 $\rightarrow 4 \times 4$

“Same”: Pad so that output size is the same as the input size.

$$n + 2p - f + 1 \times n + 2p - f + 1$$

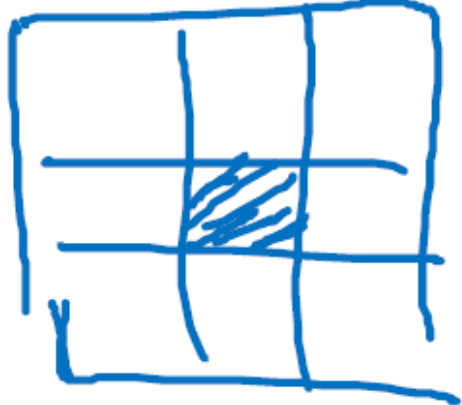
$$\cancel{n + 2p - f + 1} = \cancel{n} \Rightarrow \boxed{p = \frac{f-1}{2}}$$

$$3 \times 3 \quad p = \frac{3-1}{2} = 1$$

$$5 \times 5 \quad f=5$$

f is usually odd

1×1
 3×3
 5×5
 7×7



$p=2$

Strided convolution

2	3	7	4	6	2	9
6	6	9	8	7	4	3
3	4	8	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7×7

*

3	4	4
1	0	2
-1	0	3

3×3

Stride = 2

=

Strided convolution

2 ³	3 ⁴	7 ⁴	4	6	2	9
6 ¹	6 ⁰	9 ²	8	7	4	3
3 ⁻¹	4 ⁰	8 ³	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

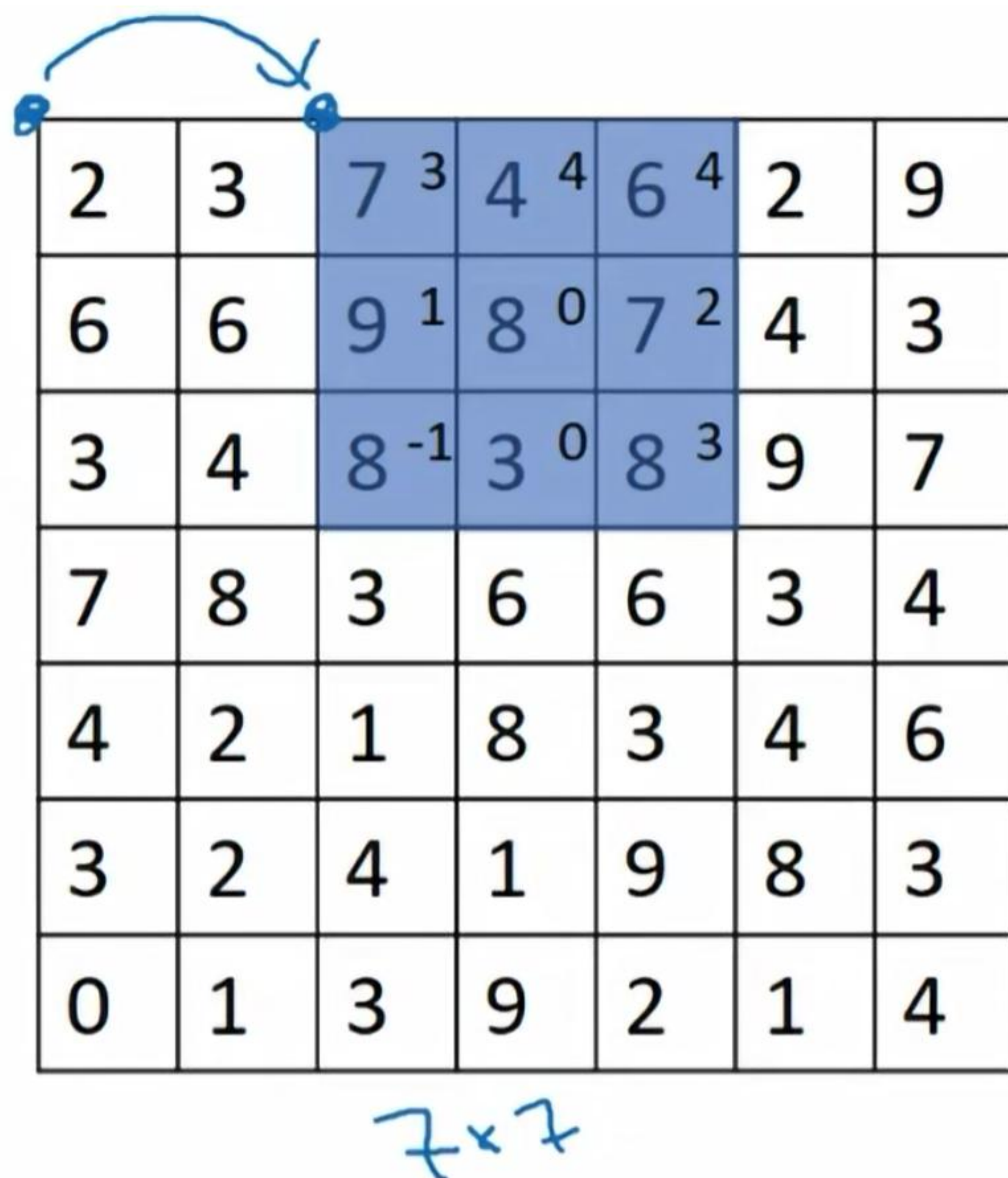
3x3

Stride = 2

=

9.1		

Strided convolution



2	3	7 ³	4 ⁴	6 ⁴	2	9
6	6	9 ¹	8 ⁰	7 ²	4	3
3	4	8 ⁻¹	3 ⁰	8 ³	9	7
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

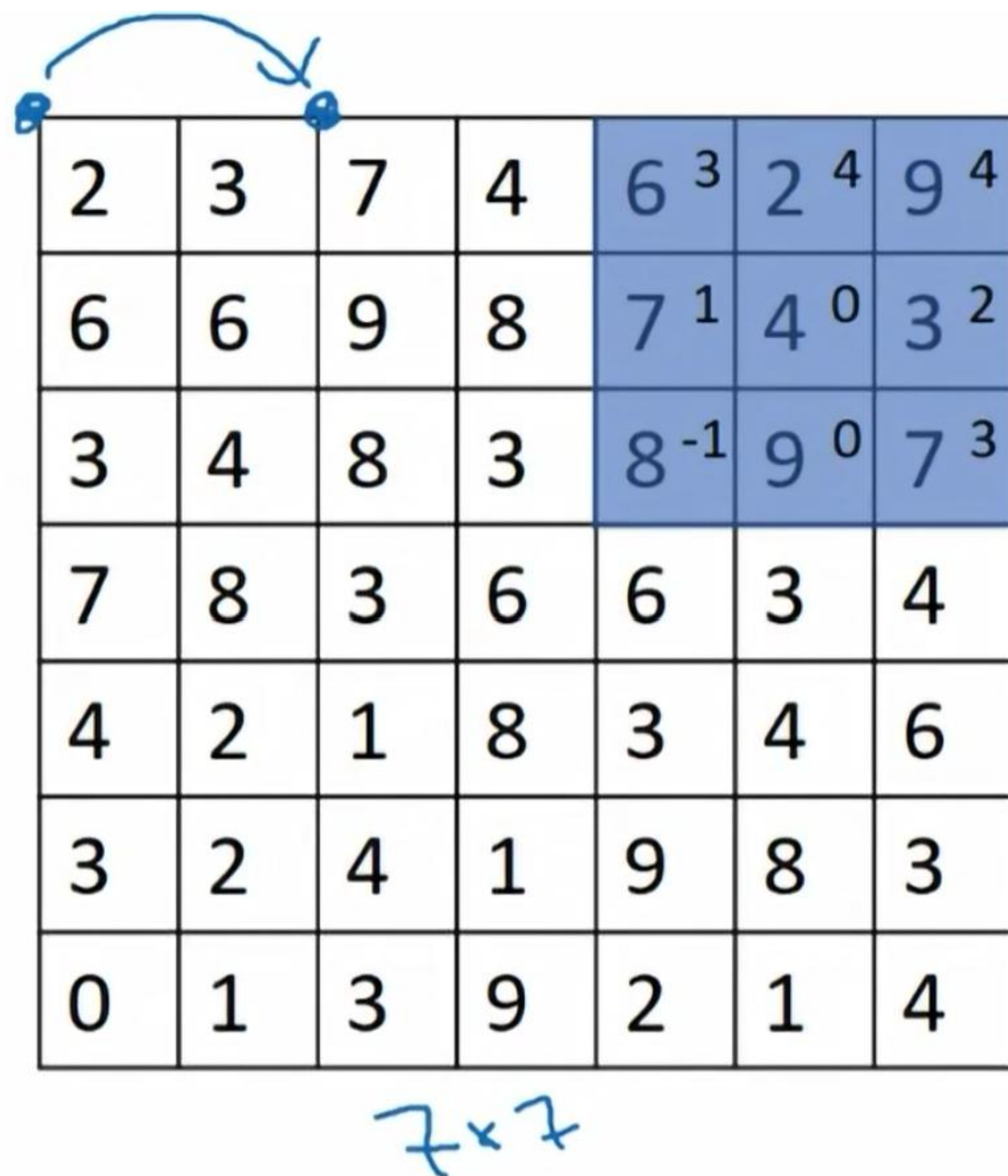
3x3

Stride = 2

=

91	100	

Strided convolution



2	3	7	4	6 ³	2 ⁴	9 ⁴
6	6	9	8	7 ¹	4 ⁰	3 ²
3	4	8	3	8 ⁻¹	9 ⁰	7 ³
7	8	3	6	6	3	4
4	2	1	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

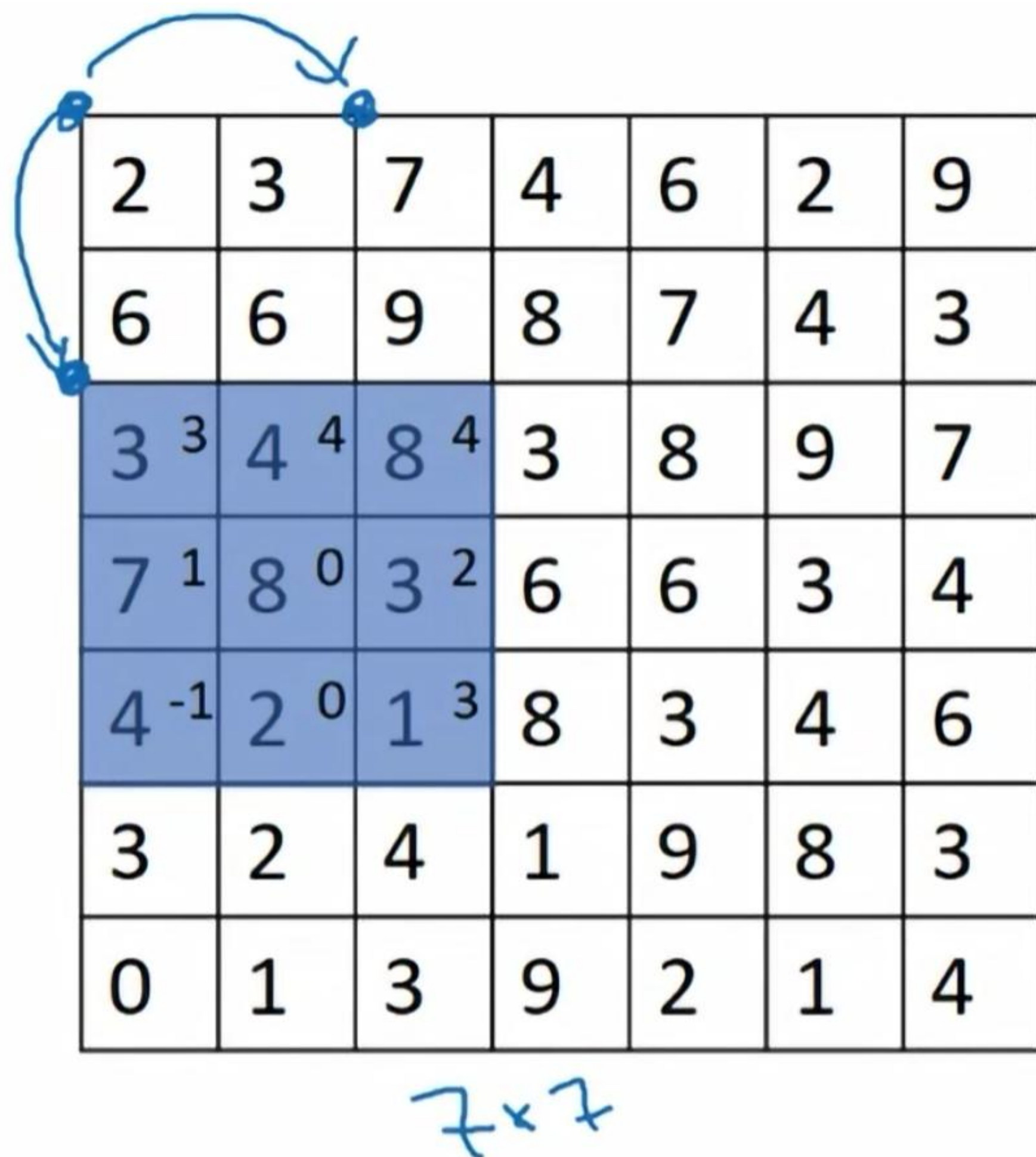
3x3

Stride = 2

=

91	100	83

Strided convolution



2	3	7	4	6	2	9
6	6	9	8	7	4	3
3 ³	4 ⁴	8 ⁴	3	8	9	7
7 ¹	8 ⁰	3 ²	6	6	3	4
4 ⁻¹	2 ⁰	1 ³	8	3	4	6
3	2	4	1	9	8	3
0	1	3	9	2	1	4

7x7

*

3	4	4
1	0	2
-1	0	3

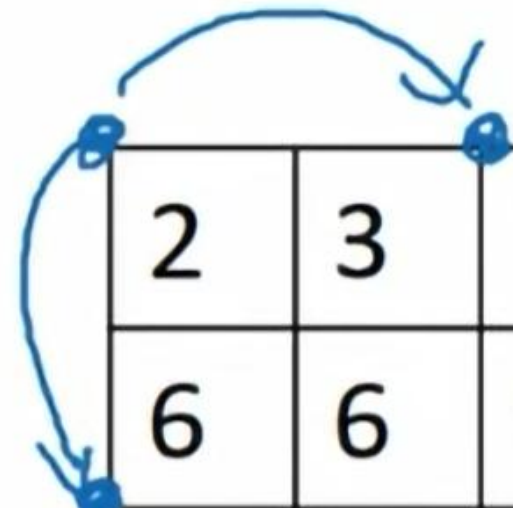
3x3

Stride = 2

=

91	100	83
69		

Strided convolution



2	3	7	4	6	2	9
6	6	9	8	7	4	3
3	4	8	3	8	9	7
7	8	3	6	6	3	4
4	2	1	8	3 ³	4 ⁴	6 ⁴
3	2	4	1	9 ¹	8 ⁰	3 ²
0	1	3	9	2 ⁻¹	1 ⁰	4 ³

7x7

$n \times n$ * $f \times f$
 padding p stride s
 $s = 2$

3	4	4
1	0	2
-1	0	3

3x3

stride = 2

=

91	100	83
69	91	127
44	72	74

$$\left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor$$

$$\frac{7 + 0 - 3}{2} + 1 = \frac{4}{2} + 1 = 3$$

Summary of convolutions

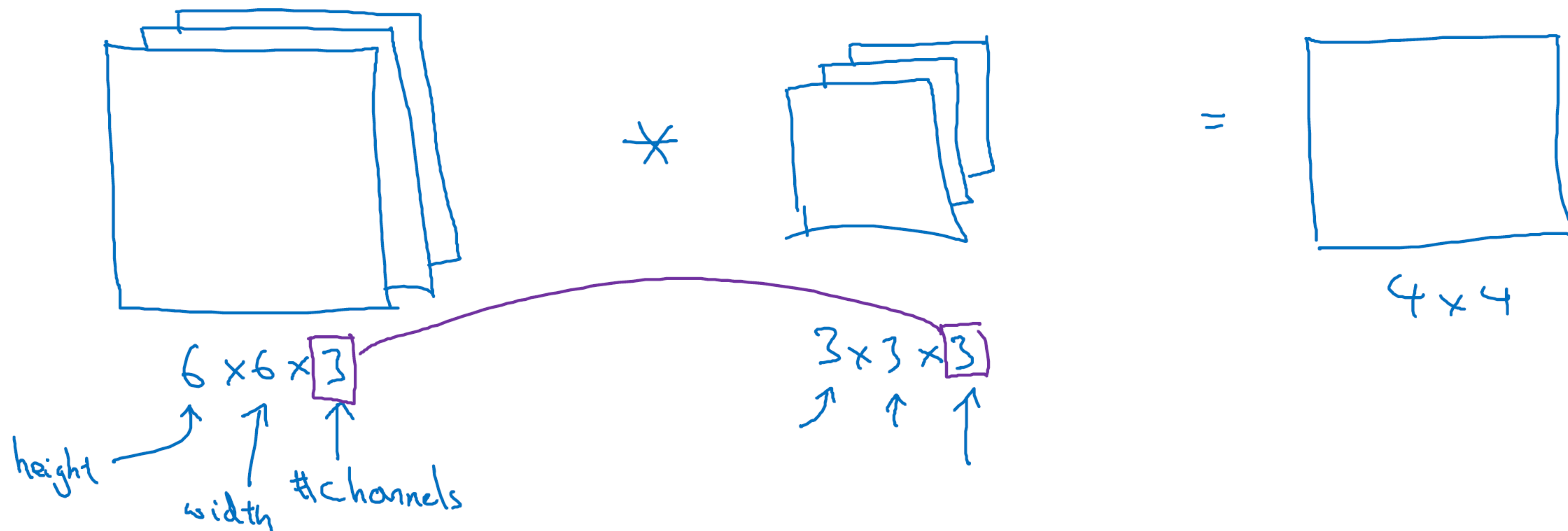
$n \times n$ image $f \times f$ filter

padding p stride s

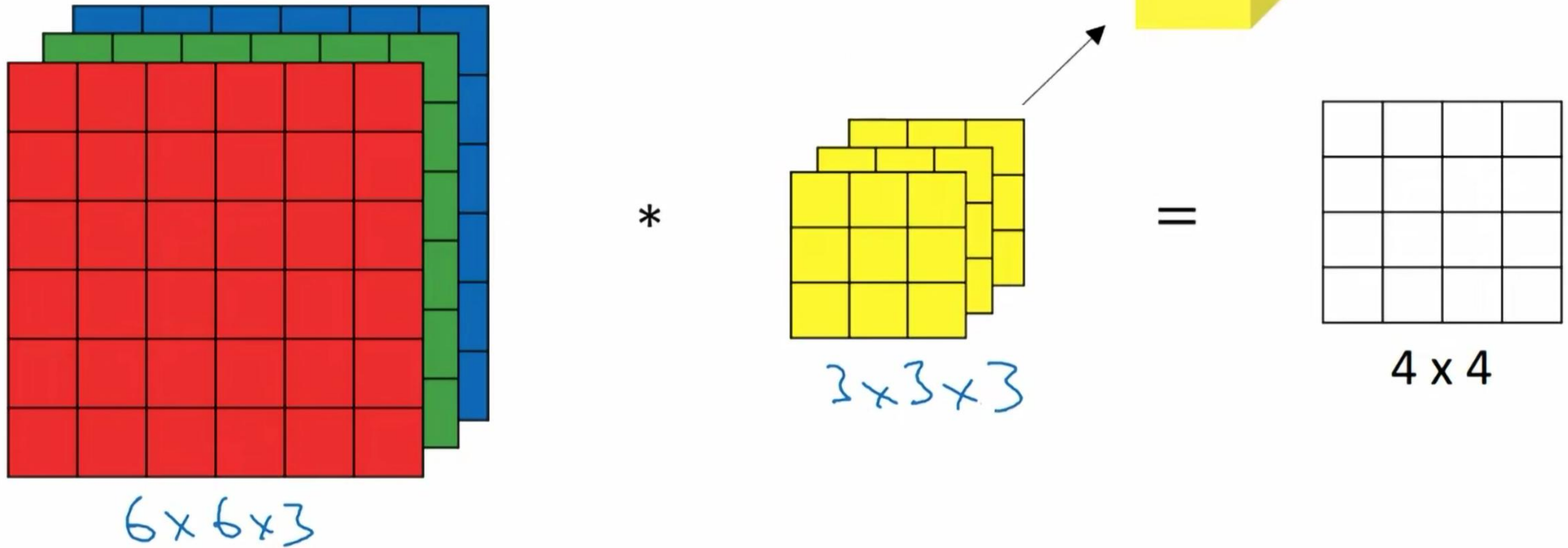
Output Size:

$$\left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n+2p-f}{s} + 1 \right\rfloor$$

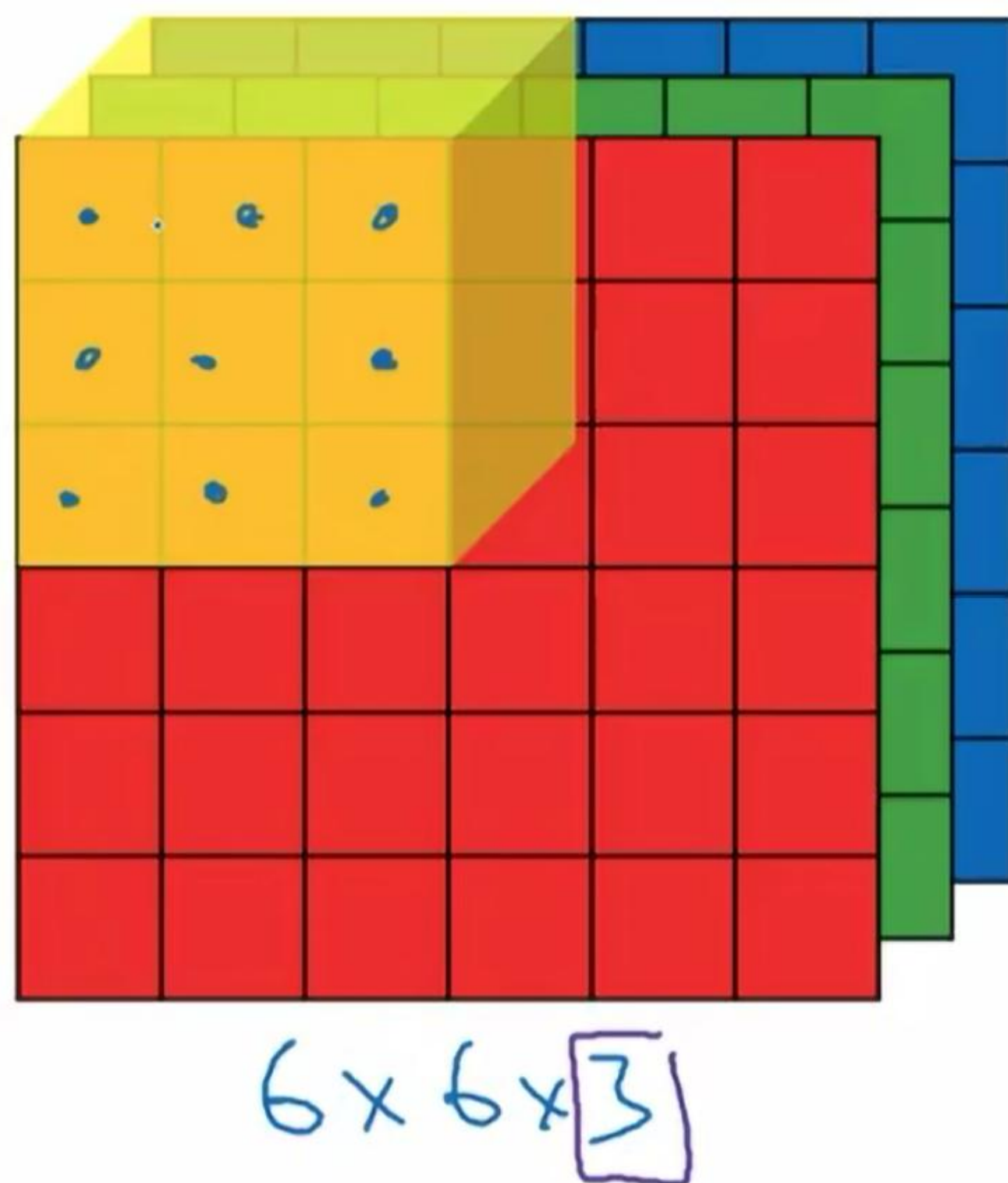
Convolutions on RGB images



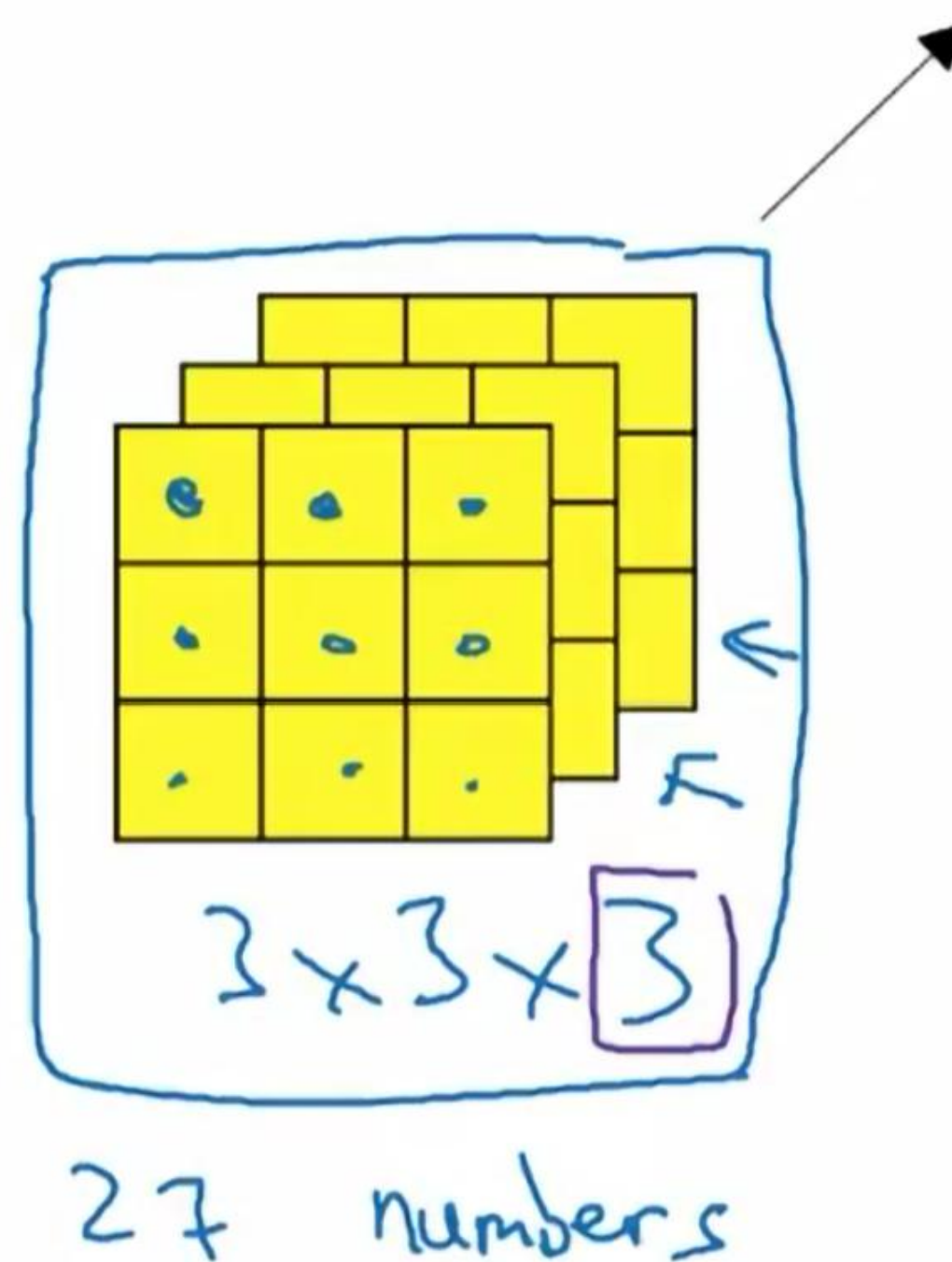
Convolutions on RGB images



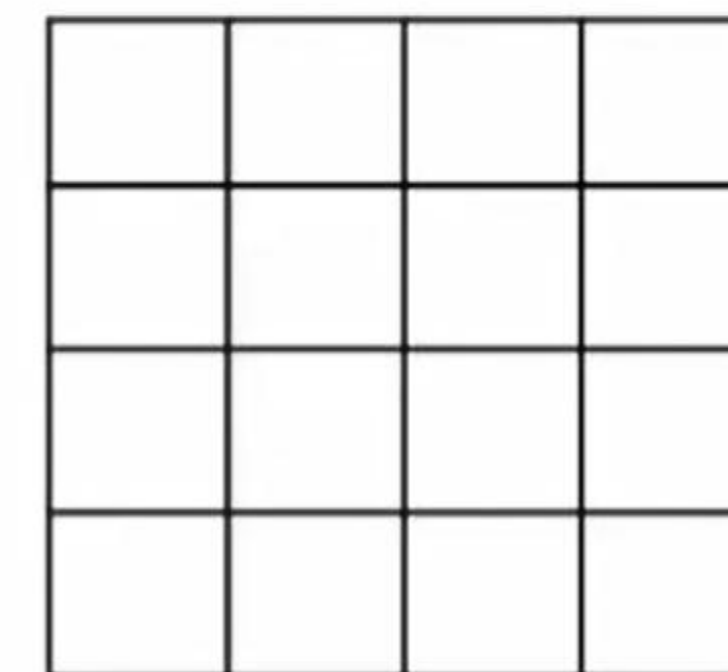
Convolutions on RGB images



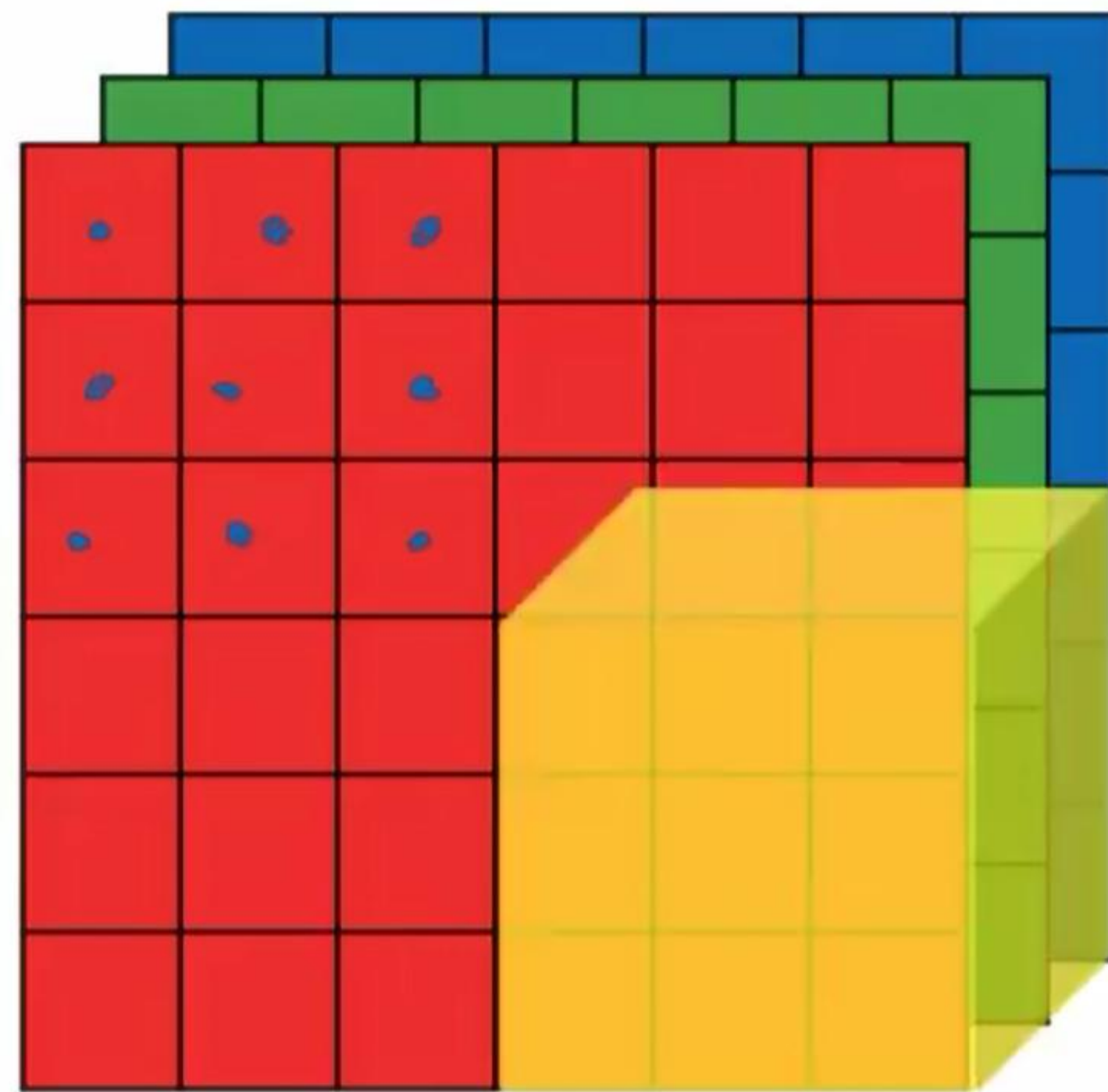
*



=

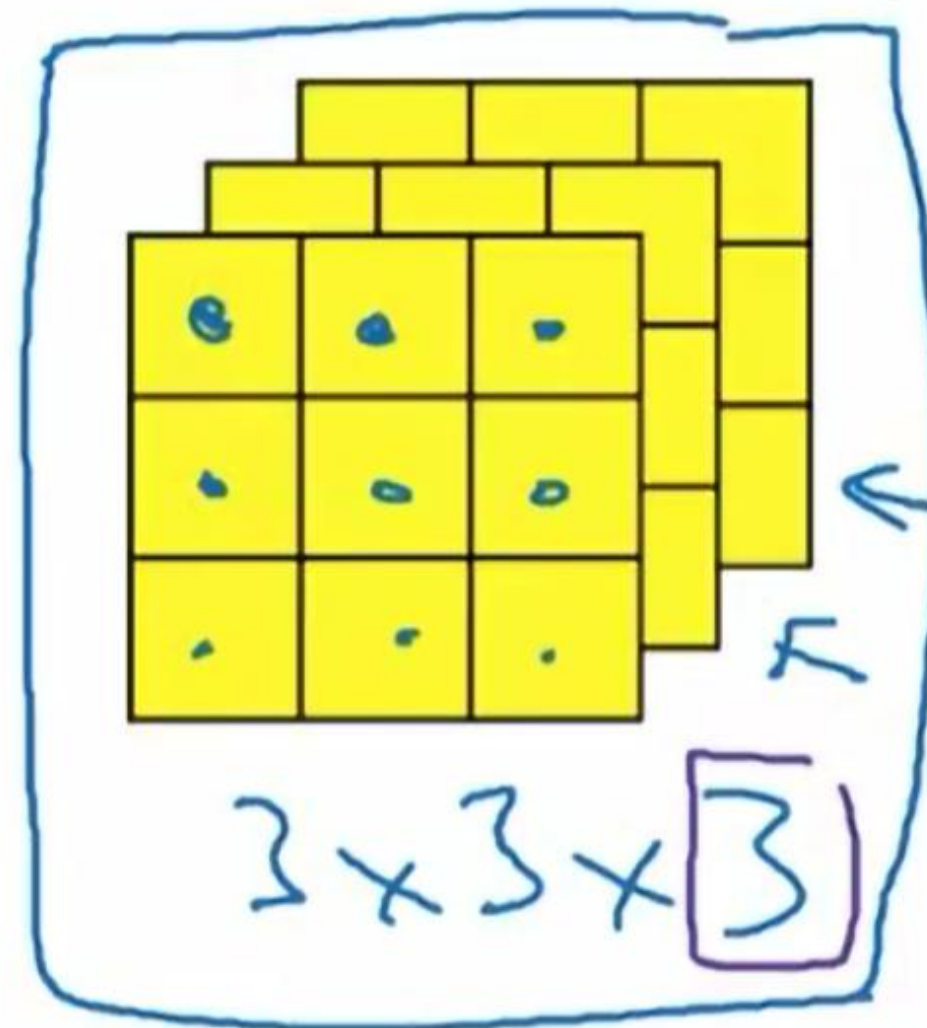
 4×4

Convolutions on RGB images

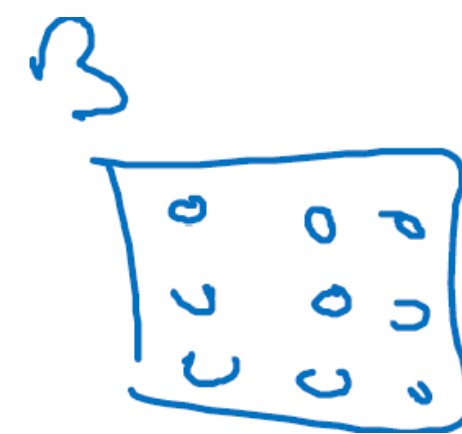
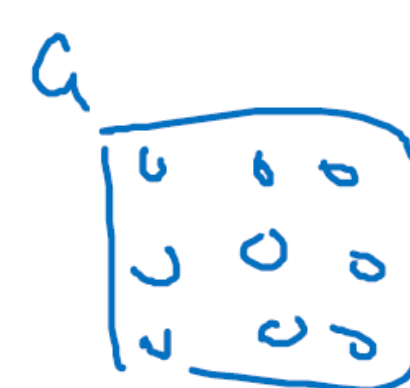
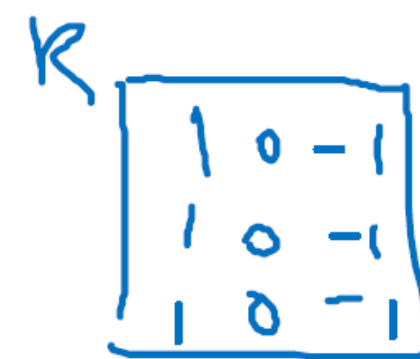


$6 \times 6 \times 3$

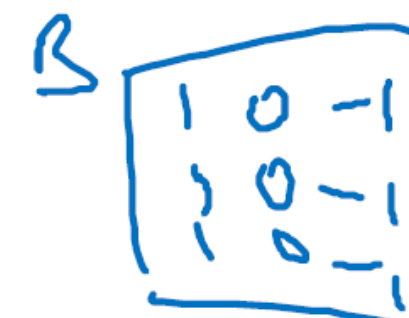
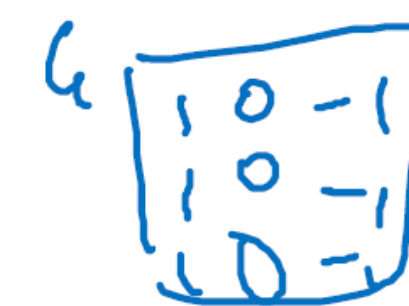
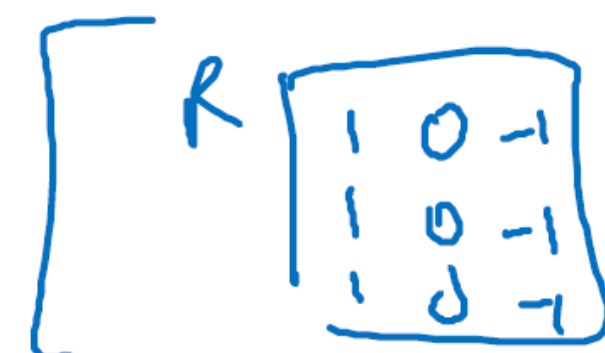
*



27 numbers

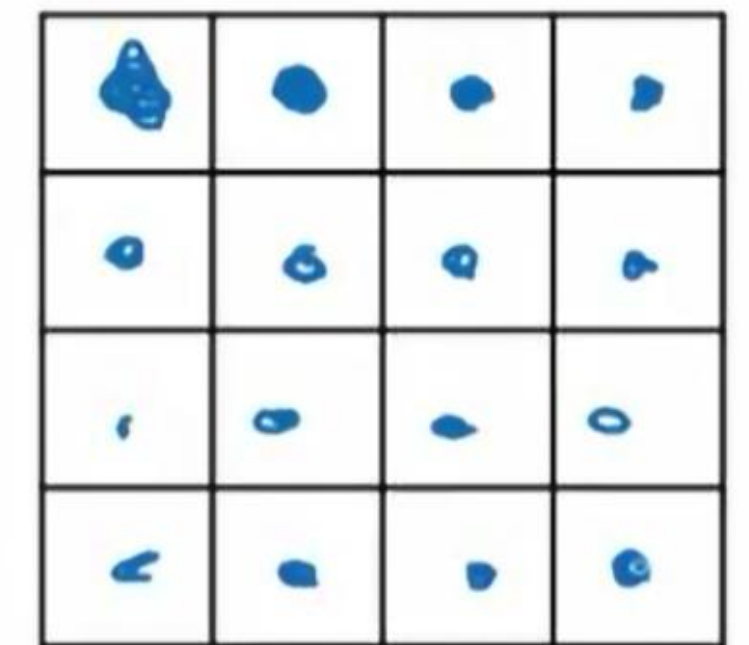


$\rightarrow 3 \times 3 \times 3$



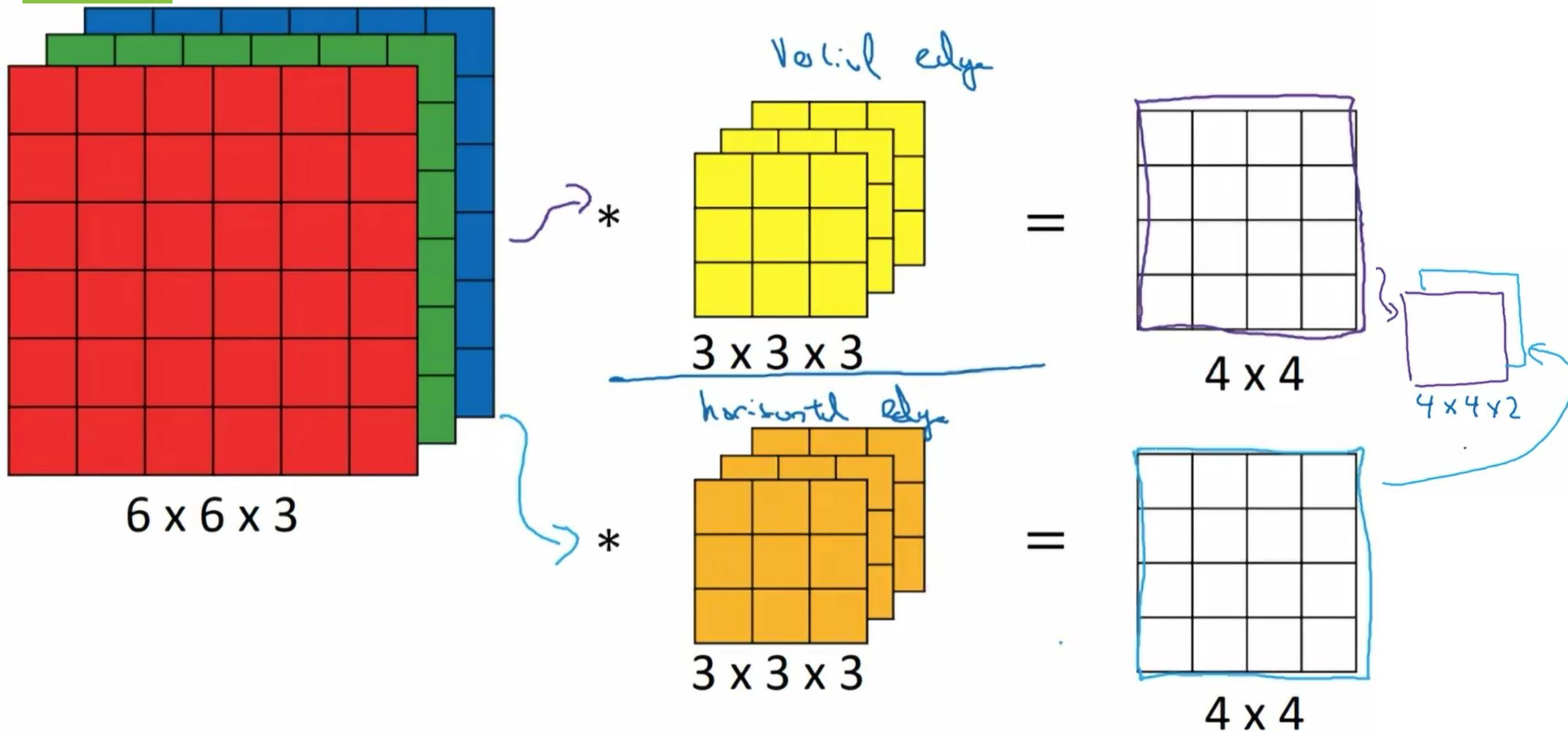
$\rightarrow 3 \times 3 \times 3$

=

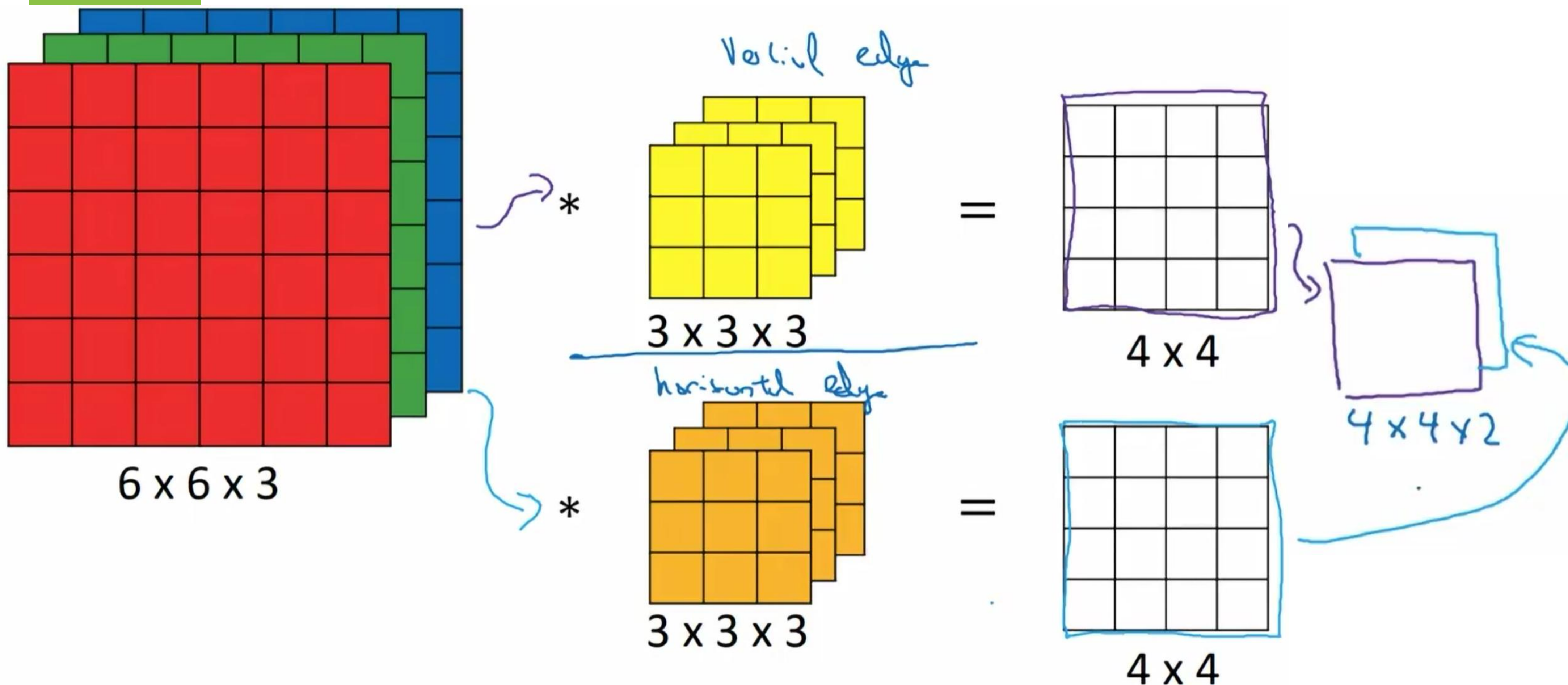


4×4

Multiple filters



Multiple filters

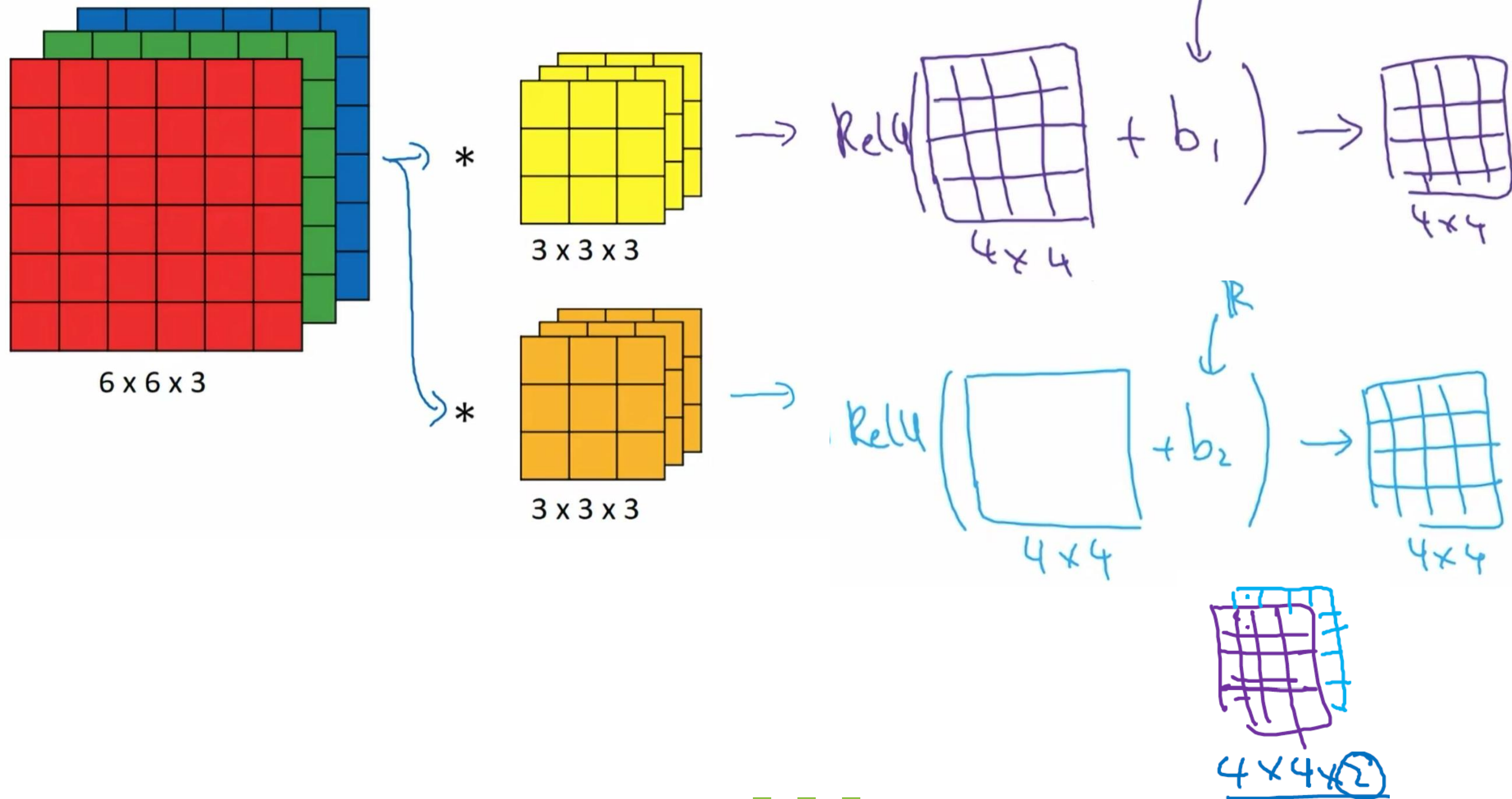


Summary: $n \times n \times n_c$
 $6 \times 6 \times 3$

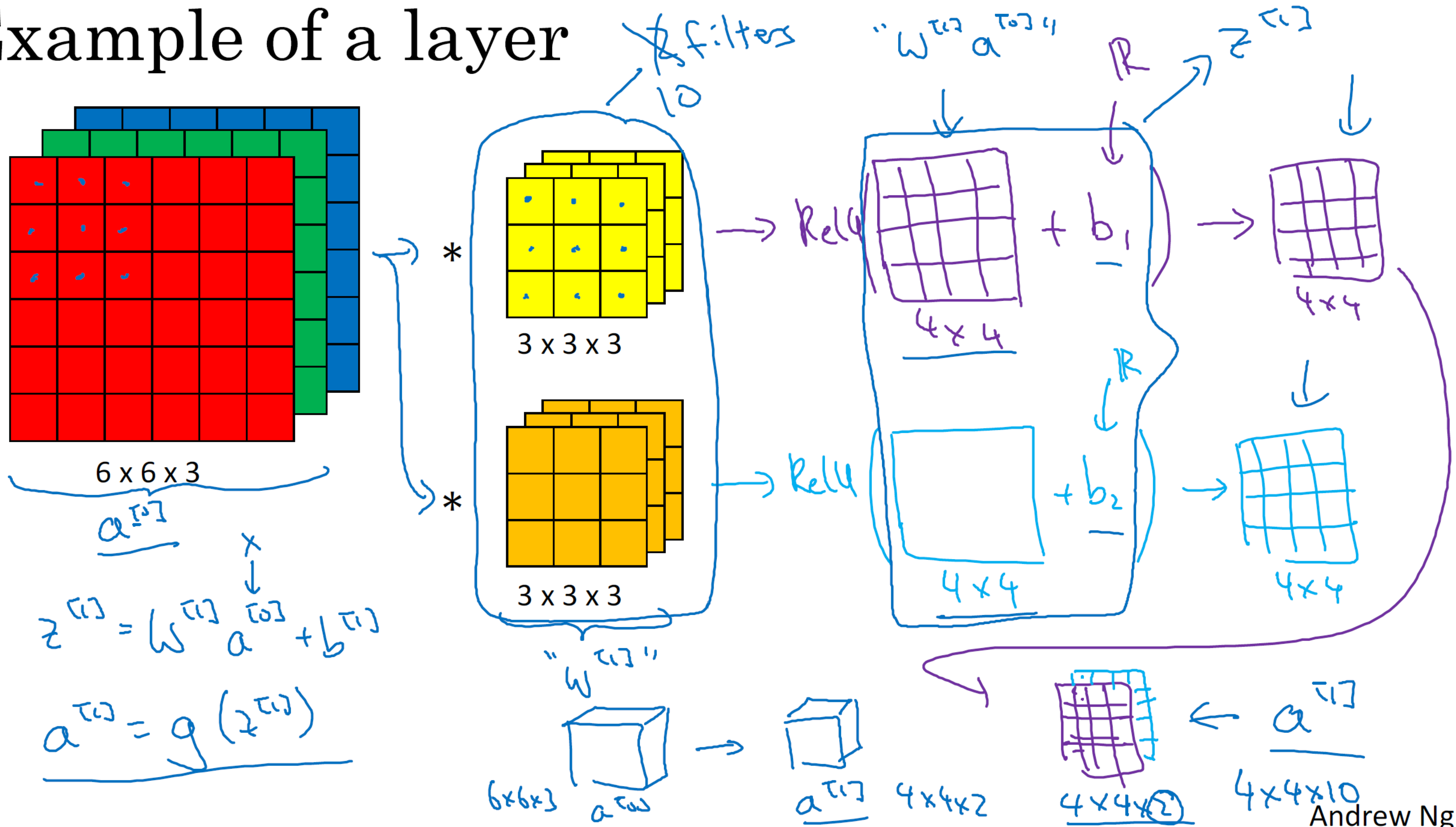
$\times f \times f \times n_c$
 $3 \times 3 \times 3$

$\rightarrow \frac{n-f+1}{4} \times \frac{n-f+1}{4} \times n_c' \uparrow \# \text{ filters}$

Example of a layer

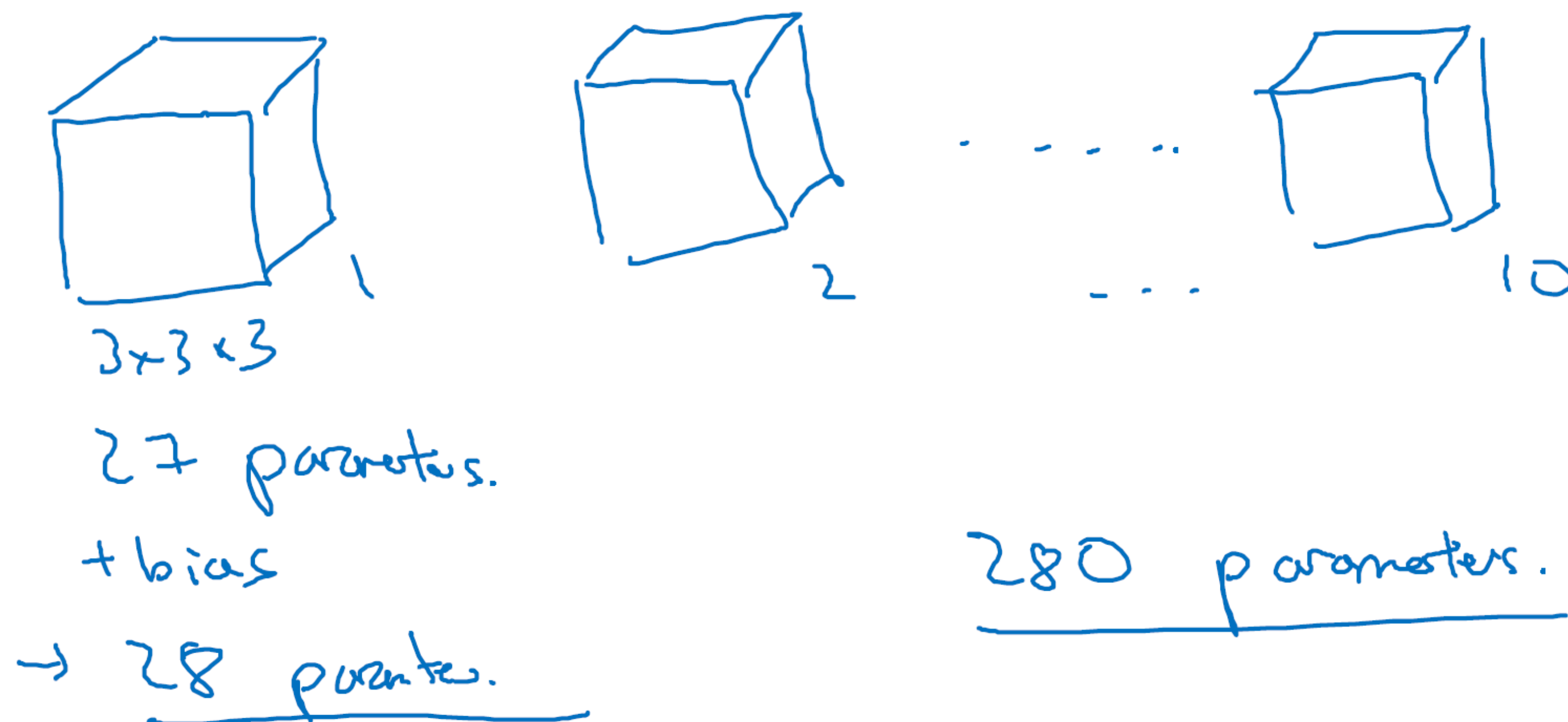


Example of a layer



Number of parameters in one layer

If you have 10 filters that are $3 \times 3 \times 3$ in one layer of a neural network, how many parameters does that layer have?



Summary of notation

If layer l is a convolution layer:

$f^{[l]}$ = filter size

$p^{[l]}$ = padding

$s^{[l]}$ = stride

$n_c^{[l]}$ = number of filters

→ Each filter is: $f^{[l]} \times f^{[l]} \times n_c^{[l-1]}$

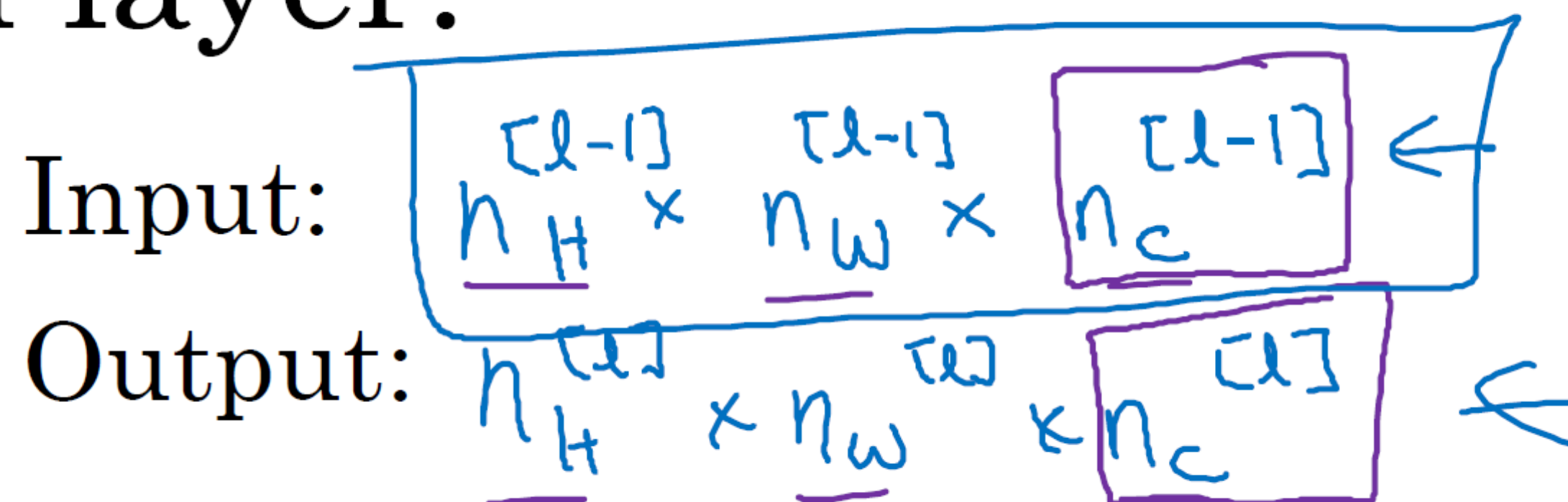
Activations: $a^{[l]} \rightarrow n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}$

Weights: $f^{[l]} \times f^{[l]} \times n_c^{[l-1]} \times n_c^{[l]}$

bias: $n_c^{[l]} - (1, 1, 1, n_c^{[l]})$ ← #filters in layer l.

Input:

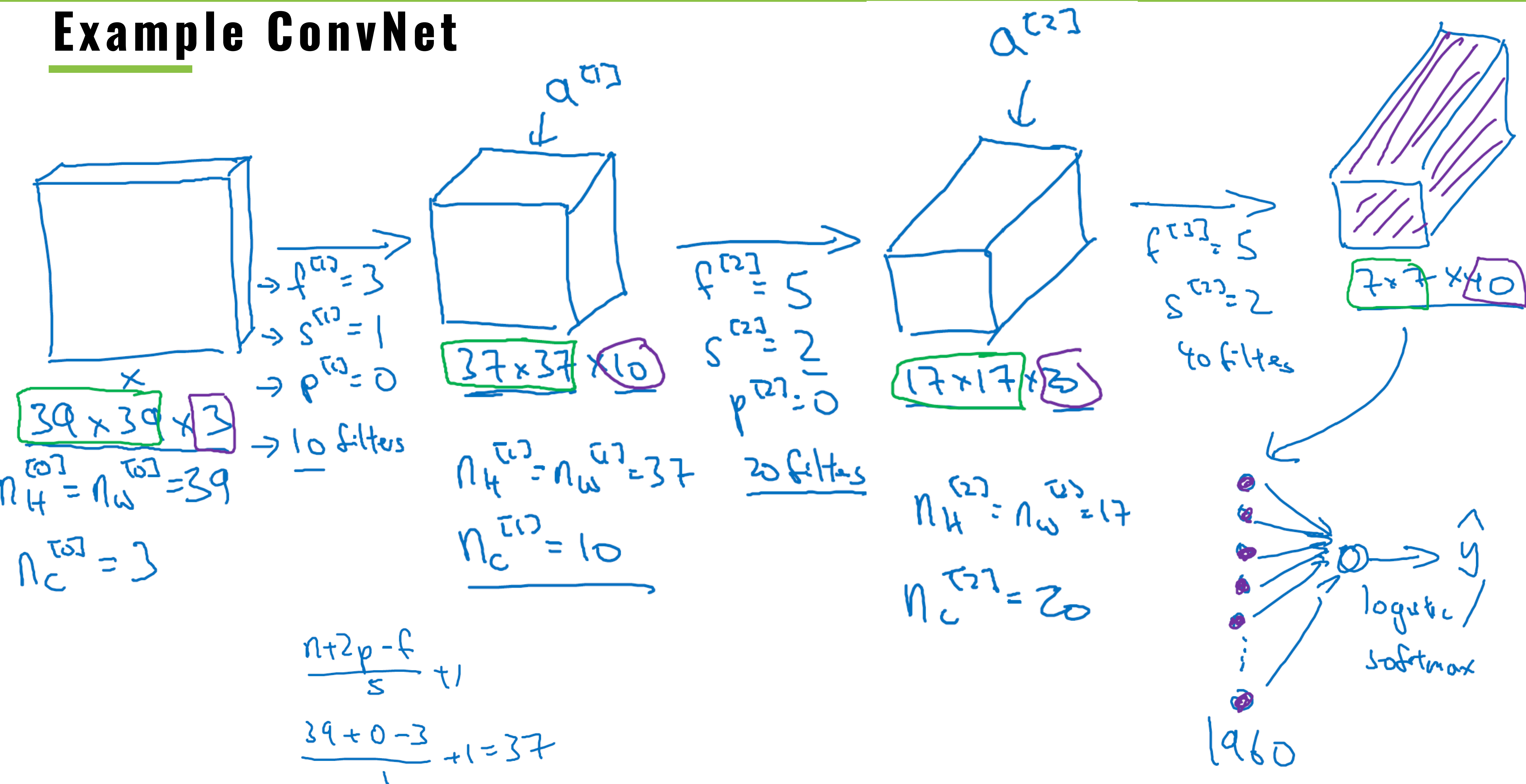
Output:



$$n_H^{[l]} = \left\lfloor \frac{n_H^{[l-1]} + 2p^{[l]} - f^{[l]}}{s^{[l]}} + 1 \right\rfloor$$

$$A^{[l]} \rightarrow m \times \underbrace{n_H^{[l]} \times n_W^{[l]} \times n_c^{[l]}}_{n_c^{[l]} \times n_H^{[l]} \times n_W^{[l]}}$$

Example ConvNet



Types of layer in a convolutional network

- Convolution (CONV) ←
 - Pooling (POOL) ←
 - Fully connected (FC) ←
- 

Pooling layer: Max pooling

1	3	2	1
2	9	1	1
1	3	2	3
5	6	1	2

4x4

9	2
6	3

2x2

Hyperparameters:
 $f = 2$
 $s = 2$

Pooling layer: Max pooling

1	3	2	1	3
2	9	1	1	5
1	3	2	3	2
8	3	5	1	0
5	6	1	2	9

5x5

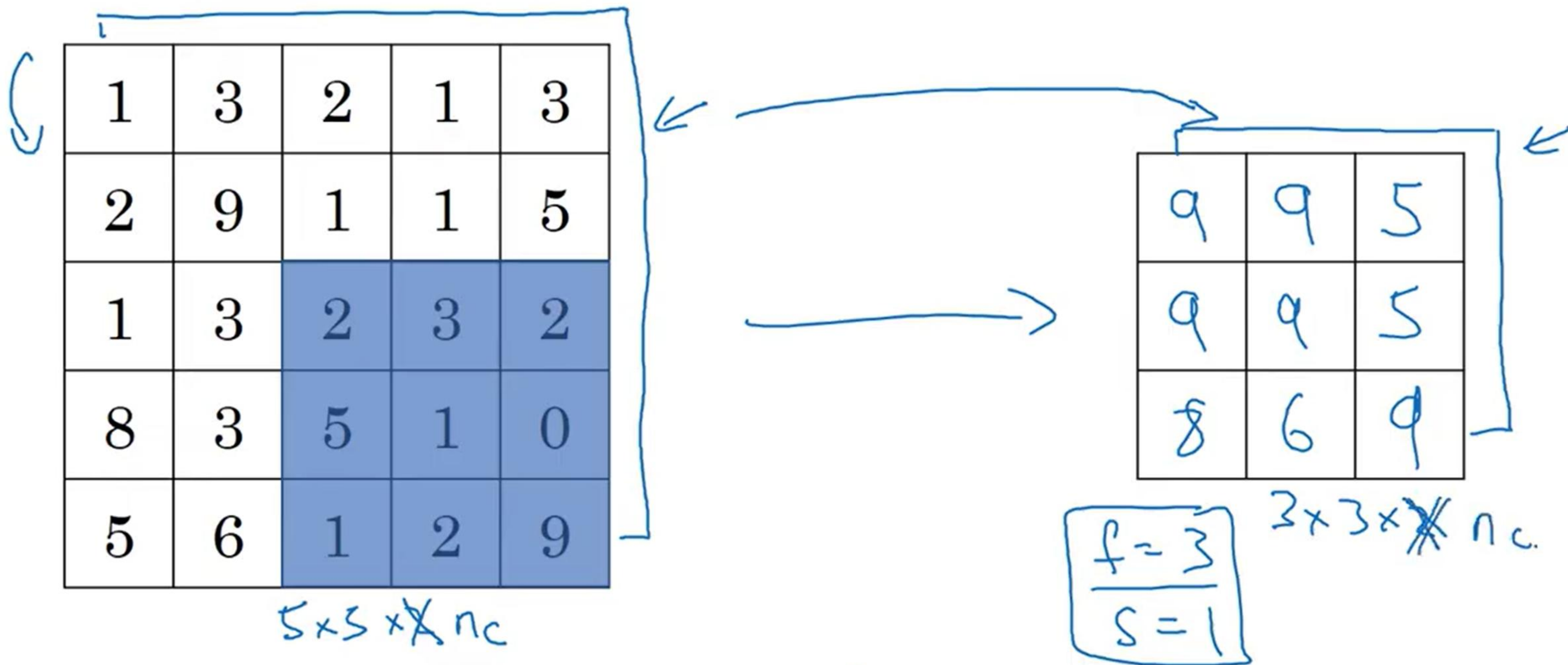
$$\left\lfloor \frac{n + 2p - f}{s} + 1 \right\rfloor$$

9	9	5
9	9	5
8	6	9

$$f = 3$$

$$s = 1$$

Pooling layer: Max pooling



Pooling layer: Average pooling

1	3	2	1
2	9	1	1
1	4	2	3
5	6	1	2



3.75	1.25
4	2

$$f=2$$

$$s=2$$

$$\underline{7 \times 7 \times 1000} \rightarrow 1 \times 1 \times 1000$$

Summary of pooling

Hyperparameters:

f : filter size

$$f=2, s=2$$

$$f=3, s=2$$

s : stride

Max or average pooling

⇒ ~~p: padding.~~

No parameters to learn!

$$n_H \times n_W \times \underline{n_C}$$

$$\downarrow$$

$$\left\lfloor \frac{n_H - f}{s} + 1 \right\rfloor \times \left\lfloor \frac{n_W - f}{s} + 1 \right\rfloor$$

$$\times \underline{n_C}$$

Thank You

Any questions?

kushol@iut-dhaka.edu

Source & Special Thanks to Andrew Ng (Coursera)

<https://www.youtube.com/@Deeplearningai>

<https://www.youtube.com/playlist?list=PLkDaE6sCZn6Gl29AoE31iwdVwSG-KnDzF>